

# 8

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## Schwarz-Christoffel Integrals

In practical applications of conformal mapping of a standard region (the half-plane or the unit disk) onto a problem region which is in the form of a polygon, it becomes necessary to determine approximately the  $(2n+2)$  parameters  $\alpha_1, \dots, \alpha_n, x_1, \dots, x_n$ , and the constants  $C_1$  and  $C_2$  that appear in the Schwarz-Christoffel formula (7.1.1). Evaluation of these quantities is known as the *parameter problem*. We have seen that the mapping functions obtained by using the Schwarz-Christoffel formula involve certain improper integrals which are known as Schwarz-Christoffel integrals. We will discuss methods for numerical solution of these integrals and present Newton's method for the general case of mapping the upper half-plane onto a quadrilateral.

### 8.1. Parameter Problem

If the values of all  $(2n+2)$  parameters in the Schwarz-Christoffel formula (7.1.1) or (7.1.6) are known, then the polygon is uniquely defined, and the coordinates  $w_k$  of the vertices are given by

$$w_k = C_1 \int_0^{x_k} \prod_{i=1}^n (\zeta - x_i)^{-\alpha_i} d\zeta + C_2, \quad k = 1, \dots, n. \quad (8.1.1)$$

The length of the side joining the vertices  $w_k$  and  $w_{k+1}$  of the polygon is determined by

$$|w_k, w_{k+1}| = |w_{k+1} - w_k| = |C_1| \int_{x_k}^{x_{k+1}} \prod_{i=1}^n (\zeta - x_i)^{-\alpha_i} d\zeta. \quad (8.1.2)$$

Thus, it is very easy to figure out the behavior of the  $(2n+2)$  parameters in the formula (7.1.1) or (7.1.6). In fact, the parameters  $\alpha_1, \dots, \alpha_n$  are related to the quantities  $\mu_1, \dots, \mu_n$  which are the ratios of the interior angles of the polygon to  $\pi$ . The parameter  $C_2$  affects the location of the vertices of the polygon. Any change in  $C_2$  changes the coordinates of every vertex by the same amount through homothety and translation and displaces the polygon as a unit. The parameter  $C_1$  which as a factor in the formula (8.1.2) affects the lengths of

the sides of the polygon. Any change in  $C_1$  changes all sides of the polygon by the same amount and rotates the polygon as a unit. Any change in the parameters  $x_1, \dots, x_n$  also produces relative change in the lengths of the sides of the polygon.

Let the ratio of the second, third,  $\dots$ ,  $(n-2)$ th side of the polygon to the first side be denoted by  $\lambda_2, \lambda_3, \dots, \lambda_{n-2}$ , respectively, i.e.,

$$\lambda_j = \frac{|w_{j+1} - w_j|}{|w_2 - w_1|}, \quad j = 2, 3, \dots, n-2. \quad (8.1.3)$$

Consider the function

$$\tilde{w} = \int_0^z \prod_{i=1}^n (\zeta - x_i)^{-\alpha_i} d\zeta. \quad (8.1.4)$$

If we choose the numbers  $x_1, \dots, x_n$  in this function such that the relations (8.1.3) are satisfied, where  $\alpha_i = 1 - \mu_i$ , the function  $\tilde{w}$  will define a conformal mapping of the upper half-plane onto a polygon  $\tilde{G}$ . In order to pass from the polygon  $\tilde{G}$  to  $G$ , we use the linear transformation

$$w = C_1 \tilde{w} + C_2, \quad (8.1.5)$$

which does the following: It transfers any vertex, say  $\tilde{w}_k$ , of the polygon  $\tilde{G}$  to the corresponding vertex  $w_k$  of the polygon  $G$ , then rotates the polygon  $\tilde{G}$  about the vertex  $\tilde{w}_k$  so that its sides become parallel to the sides of the polygon  $G$ , and finally, without changing the position of the vertex  $\tilde{w}_k$ , it changes the lengths of all sides of the polygon  $\tilde{G}$  such that the polygon  $\tilde{G}$  coincides with  $G$ . The constants  $C_1$  and  $C_2$  in (8.1.3) can be determined by comparing the location and sides of the polygons  $\tilde{G}$  and  $G$ .

In order to determine the parameters  $x_1, \dots, x_n$ , there are only  $(n-3)$  equations, since three of these points can be chosen arbitrarily. We can use a Möbius transformation to carry the upper half-plane onto itself such that the three points of the  $x$ -axis go into three preassigned points of the  $u$ -axis. Let these three preassigned points be denoted by  $p_1, p_2, p_3$ . Let us denote the improper integrals thus obtained from formula (7.1.1) by  $I_m$ ,  $m = 1, 2, \dots, n-2$ , and define them as

$$\begin{aligned} I_1 &= \int_{p_1}^{p_2} (\zeta - p_1)^{-\alpha_1} (p_2 - \zeta)^{-\alpha_2} (x_3 - \zeta)^{-\alpha_3} \dots (x_{n-1} - \zeta)^{-\alpha_{n-1}} \\ &\quad \times (p_3 - \zeta)^{-\alpha_n} d\zeta, \\ I_2 &= \int_{p_2}^{x_3} (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (x_3 - \zeta)^{-\alpha_3} \dots (x_{n-1} - \zeta)^{-\alpha_{n-1}} \\ &\quad \times (p_3 - \zeta)^{-\alpha_n} d\zeta, \\ &\vdots \\ I_{n-2} &= \int_{x_{n-2}}^{x_{n-1}} (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (\zeta - x_3)^{-\alpha_3} \dots (\zeta - x_{n-2})^{-\alpha_{n-1}} \\ &\quad \times (p_3 - \zeta)^{-\alpha_n} d\zeta. \end{aligned} \quad (8.1.6)$$

These integrals are known as *Schwarz-Christoffel integrals*. Then, in view of (8.1.3),

$$I_j(x_3, x_4, \dots, x_{n-1}) = \lambda_j I_1(x_3, x_4, \dots, x_{n-1}), \quad j = 2, 3, \dots, n-2. \quad (8.1.7)$$

In view of the Riemann mapping theorem (§2.6), once the three points  $x_1, x_2, x_3$  are chosen arbitrarily, the system of equations (8.1.7) has a unique solution.

In order to solve the system (8.1.7) numerically, we use Newton's method which is as follows: Let  $\tilde{x}_3, \tilde{x}_4, \dots, \tilde{x}_{n-1}$  denote the solution. Let us take the initial guess for these values as  $x_3^{(0)}, x_4^{(0)}, \dots, x_{n-1}^{(0)}$  which are assumed to be sufficiently close to  $\tilde{x}_3, \tilde{x}_4, \dots, \tilde{x}_{n-1}$ . If we expand each equation of the system (8.1.7) into a Taylor series in powers of the difference  $\tilde{x}_\nu - x_\nu^{(0)}$ ,  $\nu = 3, 4, \dots, n-1$ , and, as the first approximation, truncate these Taylor series after the first power of the differences, then we obtain a system of equations

$$I_j^{(0)} + \sum_{\nu=3}^{n-1} h_\nu^{(1)} \frac{\partial I_j^{(0)}}{\partial x_\nu} = \lambda_j \left[ I_1^{(0)} + \sum_{\nu=3}^{n-1} h_\nu^{(1)} \frac{\partial I_1^{(0)}}{\partial x_\nu} \right], \quad j = 2, \dots, n-2, \quad (8.1.8)$$

with a nonzero determinant, where  $h_\nu^{(1)}$ , known as the corrections of the first order, are the perturbed values of the differences  $\tilde{x}_\nu - x_\nu^{(0)}$ . Then the system (8.1.8) is reduced and solved for  $x_\nu^{(1)}$ ,  $\nu = 3, 4, \dots, n-1$ . Thus, using the initial values of  $h_\nu^{(0)}$ , the first approximations are given by

$$x_\nu^{(1)} = x_\nu^{(0)} + h_\nu^{(1)}, \quad \nu = 3, 4, \dots, n-1. \quad (8.1.9)$$

Next, the system (8.1.8) is expanded in Taylor series in powers of the differences

$$\tilde{x}_\nu - x_\nu^{(1)}, \quad \nu = 3, 4, \dots, n-1, \quad (8.1.10)$$

and these series are truncated after the first powers of these differences. Thus, a new system, analogous to (8.1.8), is constructed with unknowns  $x_\nu^{(2)}$ ,  $\nu = 3, 4, \dots, n-1$ , with the perturbed values of the differences (8.1.10), except that now the new system is computed using the known values of  $x_\nu^{(1)}$ . It yields the values of  $x_\nu^{(2)}$ , and the second approximations are given by

$$x_\nu^{(2)} = x_\nu^{(1)} + h_\nu^{(2)}, \quad \nu = 3, 4, \dots, n-1. \quad (8.1.11)$$

This process is continued until we reach arbitrarily close to the solutions  $\tilde{x}_\nu$ , such that the difference between two consecutive approximate values is within a prescribed tolerance.

Once the system (8.1.8) is solved, we are able to approximate the coordinates of the vertices of the polygon and the lengths of its sides. But in doing so, we must compute the Schwarz-Christoffel integrals of the form

$$E = \int_{x_k}^{x_{k+1}} (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (\zeta - x_3)^{-\alpha_3} \dots (\zeta - x_k)^{-\alpha_k} \\ \times (x_{k+1} - \zeta)^{-\alpha_{k+1}} \dots (p_3 - \zeta)^{-\alpha_n} d\zeta, \quad (8.1.12)$$

which are improper because the integrand of each integral becomes unbounded at two points where  $\zeta = x_k, x_{k+1}$  which are the limits of integration. These integrals exist because each  $\alpha_k > 0$ .

**8.1.1 Kantorovich's Method.** This method is used to solve the above integral  $E$ , as follows: Let the integrand in (8.1.12) be denoted by  $F(\zeta)$ . Then

$$\begin{aligned}
 F(\zeta) &= (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (\zeta - x_3)^{-\alpha_3} \cdots (\zeta - x_k)^{-\alpha_k} \\
 &\quad (x_{k+1} - \zeta)^{-\alpha_{k+1}} \cdots (p_3 - \zeta)^{-\alpha_n} \\
 &= (\zeta - x_k)^{-\alpha_k} [\phi(x_k) + \phi'(x_k)(\zeta - k_k)] \\
 &\quad + (x_{k+1} - \zeta)^{-\alpha_{k+1}} [\psi(x_{k+1}) - \psi'(x_{k+1})(x_{k+1} - \zeta)] \\
 &\quad + \left\{ F(\zeta) - (\zeta - x_k)^{-\alpha_k} [\phi(x_k) + \phi'(x_k)(\zeta - k_k)] \right. \\
 &\quad \left. - (x_{k+1} - \zeta) [\psi(x_{k+1}) - \psi'(x_{k+1})(x_{k+1} - \zeta)] \right\},
 \end{aligned} \tag{8.1.13}$$

where

$$\begin{aligned}
 \phi(\zeta) &= (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (\zeta - x_3)^{-\alpha_3} \cdots (\zeta - x_{k-1})^{-\alpha_{k-1}} \\
 &\quad (x_{k+1} - \zeta)^{-\alpha_{k+1}} \cdots (p_3 - \zeta)^{-\alpha_n},
 \end{aligned} \tag{8.1.14}$$

$$\begin{aligned}
 \psi(\zeta) &= (\zeta - p_1)^{-\alpha_1} (\zeta - p_2)^{-\alpha_2} (\zeta - x_3)^{-\alpha_3} \cdots (\zeta - x_k)^{-\alpha_k} \\
 &\quad (x_{k+2} - \zeta)^{-\alpha_{k+2}} \cdots (p_3 - \zeta)^{-\alpha_n}.
 \end{aligned} \tag{8.1.15}$$

Let  $E = E_1 + E_2$ , where

$$\begin{aligned}
 E_1 &= \int_{x_k}^{x_{k+1}} \left\{ (\zeta - x_k)^{-\alpha_k} [\phi(x_k) + \phi'(x_k)(\zeta - x_k)] \right. \\
 &\quad \left. + (x_{k+1} - \zeta)^{-\alpha_{k+1}} [\psi(x_{k+1}) - \psi'(x_{k+1})(x_{k+1} - \zeta)] \right\} d\zeta
 \end{aligned} \tag{8.1.16a}$$

$$\begin{aligned}
 E_2 &= \int_{x_k}^{x_{k+1}} \left\{ F(\zeta) - (\zeta - x_k)^{-\alpha_k} [\phi(x_k) + \phi'(x_k)(\zeta - x_k)] \right. \\
 &\quad \left. - (x_{k+1} - \zeta) [\psi(x_{k+1}) - \psi'(x_{k+1})(x_{k+1} - \zeta)] \right\} d\zeta.
 \end{aligned} \tag{8.1.16b}$$

The integral  $E_1$  can be evaluated directly in a finite form, and since  $E_2$  has no singularities it can be approximated by any formula for numerical integration of definite integrals, like Simpson's rule.

For numerical parameter problems and related methods, see §8.7.

## 8.2 Newton's Method

We will discuss Newton's method for mapping the upper half-plane onto an arbitrary quadrilateral (see Map 7.10). In fact, in order to determine  $k$  in (7.2.14), we have only one equation from (8.1.8)

$$I_4(k) = \lambda I_1(k), \tag{8.2.1}$$

where

$$\begin{aligned}
 I_4(k) &= \int_3^\infty (\zeta + 1)^{-\alpha_1} (\zeta - 1)^{-\alpha_2} (\zeta - k)^{-\alpha_3} (\zeta - 3)^{-\alpha_4} d\zeta \\
 &\quad + \int_{-1}^{-\infty} (-1 - \zeta)^{-\alpha_1} (1 - \zeta)^{-\alpha_2} (k - \zeta)^{-\alpha_3} (3 - \zeta)^{-\alpha_4} d\zeta,
 \end{aligned} \tag{8.2.2}$$

and

$$I_1(k) = \int_{-1}^1 (\zeta + 1)^{-\alpha_1} (1 - \zeta)^{-\alpha_2} (k - \zeta)^{-\alpha_3} (3 - \zeta)^{-\alpha_4} d\zeta. \quad (8.2.3)$$

In  $I_4(k)$ , set  $\zeta = 2 + 1/t$  in the first integral and  $\zeta = -1/t$  in the second. Then

$$\begin{aligned} I_4(k) = & \int_0^1 (1 + 3t)^{-\alpha_1} (1 + t)^{-\alpha_2} (1 + (2 - k)t)^{-\alpha_3} (1 - t)^{-\alpha_4} dt \\ & + \int_0^1 (1 - t)^{-\alpha_1} (1 + t)^{-\alpha_2} (1 + kt)^{-\alpha_3} (1 + 3t)^{-\alpha_4} d\zeta. \end{aligned} \quad (8.2.4)$$

Let  $F(k) = I_4(k) - \lambda I_1(k)$ . Then  $F'(k) > 0$ . To see this, we have from (8.1.13)

$$\begin{aligned} F'(k) = & \alpha_3 \int_0^1 t(1 + 3t)^{-\alpha_1} (1 + t)^{-\alpha_2} (1 + (2 - k)t)^{-\alpha_3-1} (1 - t)^{-\alpha_4} dt \\ & - \alpha_3 \int_0^1 t(1 - t)^{-\alpha_1} (1 + t)^{-\alpha_2} (1 + kt)^{-\alpha_3-1} (1 + 3t)^{-\alpha_4} dt \\ & + \lambda \alpha_3 \int_{-1}^1 (\zeta + 1)^{-\alpha_1} (1 - \zeta)^{-\alpha_2} (k - \zeta)^{-\alpha_3-1} (3 - \zeta)^{-\alpha_4} d\zeta. \end{aligned} \quad (8.2.5)$$

Note that  $F'(1)$  and  $F'(3)$  do not exist because they are not finite. Since

$$\begin{aligned} F''(k) = & \alpha_3 (1 + \alpha_3) \left\{ \int_0^1 t^2 (1 + 3t)^{-\alpha_1} (1 + t)^{-\alpha_2} \right. \\ & (1 + (2 - k)t)^{-\alpha_3-2} (1 - t)^{-\alpha_4} dt \\ & + \int_0^1 t^2 (1 - t)^{-\alpha_1} (1 + t)^{-\alpha_2} (1 + kt)^{-\alpha_3-2} (1 + 3t)^{-\alpha_4} dt \\ & \left. - \lambda \int_{-1}^1 (\zeta + 1)^{-\alpha_1} (1 - \zeta)^{-\alpha_2} (k - \zeta)^{-\alpha_3-2} (3 - \zeta)^{-\alpha_4} d\zeta \right\}, \end{aligned} \quad (8.2.6)$$

we find that  $F''(k) > 0$  for  $1 < k < 3$ , and  $F''(1) = -\infty$  and  $F''(3) = +\infty$ . Hence, as  $k$  varies from 1 to 3, both  $F(k)$  and  $F''(k)$  increase continuously from  $-\infty$  to  $+\infty$  (for details, see Kantorovich and Krylov [1958]). Hence, the function  $F(k)$  has a zero in the interval  $(1, 3)$ . Let the zero of  $F(k)$  be denoted by  $k^*$ , and that of  $F''(k)$  by  $k^{**}$ . Now we will solve the equation  $F(k) = 0$  by Newton's method. Although the initial guess for the value of  $k$  is important in this method, we will show that it can be chosen arbitrarily from below or from above in the interval  $(1, 3)$ . Let  $k_0$  denote an arbitrary initial guess for the value of  $k$ . Then

**Theorem 8.1** *In order to solve the equation  $F(k) = 0$  by Newton's method, the value of  $k_0 \in (1, 3)$  can be chosen arbitrarily, independent of the values of  $k^*$  and  $k^{**}$ .*

PROOF. There are three cases to analyze:

**Case 1:** If  $K^{**} = k^*$ , then the initial value  $k_0$  is any number in the interval  $(1, 3)$  (see Figure 8.1). In fact, let  $k_0$  be an approximation of  $k$  from below. Then  $F(k) < 0$ , and  $F''(k) < 0$  for all  $k \in (k_0, k^*)$ . Hence, there is an  $M$  such that  $F'(k) < M$  for all  $k_0 \leq k \leq k^*$ . The first correction is given by

$$\delta^{(1)} = -\frac{F(k_0)}{F'(k_0)} > 0. \quad (8.2.7)$$

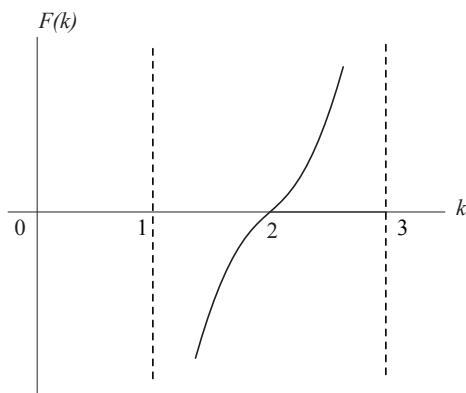


Figure 8.1 Newton's method: Case 1.

Thus, the exact first correction  $h_1$  is positive and satisfies the equation

$$F(k_0) + h_1 F'(k_0) + \frac{h_1^2}{2} F''(\tilde{k}) = 0, \quad k_0 < \tilde{k} < k^*. \quad (8.2.8)$$

Hence, from (8.2.7) and (8.2.8), we have

$$\left[ h_1 - \delta^{(1)} \right] F'(k_0) = -\frac{h_1^2}{2} F''(\tilde{k}),$$

which yields

$$h_2 = h_1 - \delta^{(1)} = -\frac{h_1^2}{2} \frac{F''(\tilde{k})}{F'(k_0)}, \quad (8.2.9)$$

which is positive in view of (8.2.8). Thus, the first approximation  $k_1$ , like  $k_0$ , is an approximation from below, and, therefore, all subsequent approximations will be very small. It is the basic property of Newton's method that the exact corrections  $h$  are always positive and decreasing. In fact, we can show that if the difference between  $h_n$  and  $h_{n+1}$  is sufficiently small, then  $h_n$  is itself very small. Assume that

$$h_n - h_{n+1} < \frac{K}{M} \varepsilon, \quad (8.2.10)$$

where  $\varepsilon > 0$  is arbitrarily small. Since  $h_n - h_{n+1} = \delta^{(n)}$ , we have  $\delta^{(n)} < \frac{K}{M} \varepsilon$ . Also,

$$\delta^{(n)} = -\frac{F(k_{n-1})}{F'(k_{n-1})} = \frac{|F(k_{n-1})|}{F'(\tilde{k})},$$

or

$$|F(k_{n-1})| = \delta^{(n)} F'(\tilde{k}). \quad (8.2.11)$$

But  $h_n$  can be evaluated from the equation

$$F(k_{n-1}) + h_n F'(\tilde{k}), \quad k_{n-1} < \tilde{k} < k^*. \quad (8.2.12)$$

Hence, from (8.2.11) and (8.2.12)

$$h_n = \frac{|F(k_{n-1})|}{F'(\tilde{k})} = \frac{\delta^{(n)} F'(\tilde{k})}{F'(\tilde{k})}, \quad h_n < \frac{K}{M} \varepsilon \frac{M}{K} = \varepsilon.$$

This shows that the sequence  $\{h_1, h_2, \dots, h_n, \dots\} \searrow 0$ . This analysis leads to the same conclusion if  $k_0$  is an approximation from above.

**Case 2:** If  $k^{**} > k^*$ , then for any initial guess  $k_0 < k^*$ , the convergence is the same as in case 1. Also,  $F(k) < 0$ , and  $F'(k) < 0$  for  $k_0 < k < k^*$  (as in case 1). Now, let  $k_0$  be such that  $k^* < k_0 < k^{**}$ . Then  $F(k_0) > 0$ , and  $F''(k) < 0$  for all  $k \leq k_0$ . Moreover,  $\delta^{(1)} < 0$ , and  $h_1 < 0$ . Now,

$$h_2 = -\frac{1}{2} h_1^2 \frac{F''(\tilde{k})}{F'(k_0)} F''(\tilde{k}) < 0,$$

and thus,  $h_2 > 0$ . This means that the first approximation is an approximation from below, and all subsequent approximations will converge from below to  $k^*$  (as in case 1). In the case when  $k_0 > k^{**}$ , we have  $F(k) > 0$  for all  $k \in (k^*, k_0)$ ;  $F'(k) \begin{cases} < 0 & \text{for all } k \in (k^*, k^{**}) \\ > 0 & \text{for all } k \in (k^{**}, k_0) \end{cases}$ ;  $\delta^{(1)} < 0$ , and  $h_1 < 0$ . Thus,  $k_1$  can be in any one of the three intervals  $(1, k^*)$ ,  $(k^*, k^{**})$ , and  $(k^{**}, k_0)$ . If  $k_1$  is in the first two intervals, then we have convergence from below to  $k^*$ . But if  $k_1$  is in the third interval, then all approximations, although decreasing continuously, still remain greater than  $k^{**}$  and approach some limiting value  $k_1 \geq k^{**}$ . The difference between  $h_n$  and  $h_{n+1}$  would be sufficiently small for sufficiently large  $n$ , and  $h_n - h_{n+1} < \frac{K}{M} \frac{k^{**} - k^*}{2}$ , where  $M = \max_{k \in (k^*, k_0)} F'(k)$ . As in case 1, it can be shown that  $|h_n| < \frac{k^{**} - k^*}{2}$ . But, by assumption, we have  $|h_n| \geq k^{**} - k^*$ . This contradiction shows that if Newton's method starts with  $k_0 > k^{**}$ , then the approximation will cross  $k^{**}$  and fall in the interval where the convergence is established as in case 1.

**Case 3:** If  $k^{**} < k^*$ , then this case can be analyzed by taking  $k_0$  in any one of the intervals discussed in cases 1 and 2. We will apply this method in Case 8.3 in the next section.

### 8.3 Numerical Computations

Now we will solve some parameter problems.

**Case 8.1.** Consider the integral in the denominator in (7.2.3). Denoting it by  $E$ , we have  $C_1 = 1/E$ , where

$$E = \left( \int_0^{1/2} + \int_{1/2}^1 \right) \zeta^{-1/2} (1 - \zeta)^{-3/4} d\zeta = E_1 + E_2.$$

Then, by (8.1.16),

$$\begin{aligned} E_1 &= \lim_{t \rightarrow 0} \int_t^{1/2} \zeta^{-1/2} (1 - \zeta)^{-3/4} d\zeta \\ &= \lim_{t \rightarrow 0} \int_t^{1/2} \left\{ \zeta^{-1/2} \left( 1 + \frac{3}{4}\zeta \right) + \zeta^{-1/2} \left[ (1 - \zeta)^{-3/4} - 1 - \frac{3}{4}\zeta \right] \right\} d\zeta \\ &= E_{11} + E_{12} \approx 1.59099 + 0.0708022 = 1.66179; \\ E_2 &= \lim_{t \rightarrow 1} \int_{1/2}^t \zeta^{-1/2} (1 - \zeta)^{-3/4} d\zeta = \lim_{t \rightarrow 1} \int_{1/2}^t \left\{ (1 - \zeta)^{-3/4} \left[ 1 + \frac{1}{2}(1 - \zeta) \right] \right. \\ &\quad \left. + (1 - \zeta)^{-3/4} \left[ \zeta^{-1/2} - 1 - \frac{1}{2}(1 - \zeta) \right] \right\} d\zeta \end{aligned}$$

$$= E_{21} + E_{22} \approx 3.53176 + 0.0505577 = 3.58232.$$

Hence  $E = E_1 + E_2 \approx 5.24412$ , and the constant  $C_1 = 1/E \approx 0.19069$ . Note that the exact value of  $E = \frac{\sqrt{\pi}\Gamma(1/4)}{\Gamma(3/4)} \approx 0.19068994$ . The hypotenuse of the triangle is given by

$$|w_3 - w_2| = C_1 \int_1^\infty \zeta^{-1/2}(1 - \zeta)^{-3/4} d\zeta \approx 1.41421356 \approx \sqrt{2}$$

with an error of  $O(10^{-10})$ . ■

**Case 8.2.** In order to determine  $k$  in (7.2.10) such that the ratio  $\frac{w_2 - w_1}{w_3 - w_2} = 2$ , note that the system (8.1.7) reduces to only one equation  $I_2 = 2I_1$ , i.e.,

$$\int_{-1}^1 (1 - \zeta^2)^{-1/2} (1 - k^2 \zeta^2)^{-1/2} d\zeta = 2 \int_1^{1/k} (\zeta^2 - 1)^{-1/2} (1 - k^2 \zeta^2)^{-1/2} d\zeta, \quad (8.3.1)$$

which is solved by Newton's method as follows: We have only one correction which we will denote by  $h$  with subscript to denote the appropriate number of approximation. Let us take the initial guess  $k = 1/2$ , and determine  $h_1$  from (8.1.8) which is

$$I_2\left(\frac{1}{2}\right) + h_1 \frac{dI_2\left(\frac{1}{2}\right)}{dk} = I_1\left(\frac{1}{2}\right) + h_1 \frac{dI_1\left(\frac{1}{2}\right)}{dk}. \quad (8.3.2)$$

The free term  $I_2(1/2) = \int_0^1 (1 - \zeta)^{-1/2} \psi(\zeta) d\zeta$ , where  $\psi(\zeta) = (1 + \zeta)^{-1/2} + (1 - \zeta^2/4)^{-1/2}$ , which yields  $I_2(1/2) \approx 2.15652$  on integration. Also,

$$\begin{aligned} \frac{dI_2\left(\frac{1}{2}\right)}{dk} &= \frac{1}{2} \int_0^1 \frac{\zeta^2}{(1 - \zeta^2)^{1/2}} \left(1 - \frac{\zeta^2}{4}\right)^{3/2} d\zeta \approx 0.541732, \\ \frac{dI_1(1/2)}{dk} &\approx -1.79181. \end{aligned}$$

Substituting these values in (8.3.2) we find that  $h_1 \approx 0.201739$ . Hence, the first approximation for  $k$  is  $k \approx 0.5 + 0.201739 \approx 0.7$ . Now, for the second approximation, first we compute  $h_2$  from (8.1.8), i.e.,

$$I_2(0.7) + h_2 \frac{dI_2(0.7)}{dk} = I_1(0.7) + h_2 \frac{dI_1(0.7)}{dk}, \quad (8.3.3)$$

which gives  $h_2 \approx 0.00668985$ , and the second approximation for  $k$  is  $k \approx 0.7 + 0.00668985 \approx 0.70669$ . However, the exact value of  $k$  can be determined in this case by setting  $\zeta = \frac{1}{\sqrt{1 - (1 - k^2)x^2}}$  in  $I_2(k)$ . Then

$$I_2(k) = \int_0^1 (1 - x^2)^{-1/2} (1 - k'^2 x^2)^{-1/2} dx,$$



where  $k'^2 = 1 - k^2$ . Hence, Eq (8.3.1) gives

$$\int_0^1 (1 - \zeta^2)^{-1/2} (1 - k^2 \zeta^2)^{-1/2} d\zeta = \int_0^1 (1 - x^2)^{-1/2} (1 - k'^2 x^2)^{-1/2} dx.$$

Thus,  $k = k'$ , which gives  $k = 1/\sqrt{2} \approx 0.7071$ . A comparison of this value of  $k$  with the second approximation for  $k$  shows that the error is about 0.04%. ■

**Case 8.3.** The Schwarz-Christoffel transformation that maps the trapezoid  $A_1 A_2 A_3 A_4$  in the  $z$ -plane onto the upper half-plane is given by the Map 7.8 (Figure 7.7). The mapping function is (7.2.21), which we rewrite here as

$$w = C_1 \int_0^z (\zeta + 1)^{-1/6} (\zeta - 1)^{-1/3} (\zeta + k)^{-2/3} (\zeta - 3)^{-5/6} d\zeta + C_2. \quad (8.3.4)$$

In order to determine the value  $k$ , we have from Eq (8.1.8)

$$I_4(k) = 3 I_1(k), \quad (8.3.5)$$

where from (8.2.4)

$$\begin{aligned} I_4(k) &= \int_0^1 (1 + 3t)^{-1/6} (1 + t)^{-1/3} (1 + (2 - k)t)^{-2/3} (1 - t)^{-5/6} dt \\ &\quad + \int_0^1 (1 - t)^{-1/6} (1 + t)^{-1/3} (1 + kt)^{-2/3} (1 + 3t)^{-5/6} dt \\ &= I_{41}(k) + I_{42}(k), \end{aligned} \quad (8.3.6)$$

and from (8.2.3)

$$\begin{aligned} I_1(k) &= \int_{-1}^1 (\zeta + 1)^{-1/6} (1 - \zeta)^{-1/3} (k - \zeta)^{-2/3} (3 - \zeta)^{-5/6} d\zeta \\ &= \left( \int_{-1}^0 + \int_0^1 \right) (\zeta + 1)^{-1/6} (1 - \zeta)^{-1/3} (k - \zeta)^{-2/3} (3 - \zeta)^{-5/6} d\zeta \\ &= I_{11}(k) + I_{12}(k). \end{aligned}$$

Let  $k = 2$  be the initial guess. Then we find that  $\delta^{(1)} \approx -0.650694$ ,  $k_1 \approx 1.34931$ ,  $\delta^{(2)} \approx 0.0150829$ ,  $k_2 \approx 1.36439$ ,  $\delta^{(3)} \approx 0.0005$ , and  $k_3 \approx 1.36554$ . Now, in order to compute  $C_2$  and  $C_1$ , note that the function

$$w = \int_0^z (\zeta + 1)^{-1/6} (\zeta - 1)^{-1/3} (\zeta + k)^{-2/3} (\zeta - 3)^{-5/6} d\zeta \quad (8.3.7)$$

maps the upper half-plane onto a trapezoid  $A_1^* A_2^* A_3^* A_4^*$  similar to the given  $A_1 A_2 A_3 A_4$ . To determine the complex coordinate  $w_1^*$  of the vertex  $A_1^*$  which corresponds to  $z = -1$ , we have

$$\begin{aligned} w_1^* &= \int_0^{-1} (\zeta + 1)^{-1/6} (\zeta - 1)^{-1/3} (\zeta + k)^{-2/3} (\zeta - 3)^{-5/6} d\zeta \\ &\approx -(-1)^{1/6} (0.24631) = -0.24631 i. \end{aligned}$$

Similarly,  $w_2^* \approx 0.90311 i$ . Hence  $w = C_2 + C_1 w^*$  yields

$$-\frac{1}{2} = C_2 - 0.24631 i C_1, \quad \frac{1}{2} = C_2 + 0.90311 i C_1,$$

which gives  $C_2 = -0.2851$ , and  $C_1 = -0.87 i$ , and the required transformation is given by

$$w \approx -0.2851 - 0.87 i \int_0^z (\zeta + 1)^{-1/6} (\zeta - 1)^{-1/3} (\zeta + k)^{-2/3} (\zeta - 3)^{-5/6} d\zeta. \blacksquare \quad (8.3.8)$$

**Map 8.1.** (Schwarz-Christoffel integral for the unit disk) In view of (7.1.4), the integral in the Schwarz-Christoffel formula (7.1.1) is approximately equal to  $\zeta^{-2}$  when  $\zeta$  is close to infinity, whereas the integrand in the formula (7.1.5) is approximately  $\zeta^{-\alpha_n}$ . These quantities are significant when the region is infinite. However, we can avoid an infinite region by mapping the upper half-plane  $\Im\{z\} > 0$  onto the unit disk by the chain of mappings  $z_1 = \frac{z-i}{z+i}$  and  $z = i \frac{1+z_1}{1-z_1}$ , where  $dz = \frac{2i}{(1-z_1)^2} dz_1$ . Then the integrand in formula (7.1.1) becomes

$$\begin{aligned} \prod_{j=1}^n (z - x_j)^{-\alpha_j} dz &= \prod_{j=1}^n \left( i \frac{1+z_1}{1-z_1} - x_j \right)^{-\alpha_j} \frac{2i}{(1-z_1)^2} dz_1 \\ &= 2i \prod_{j=1}^n \left( \frac{z_1(x_j+i) - (x_j-i)}{1-z_1} \right)^{-\alpha_j} \frac{1}{(1-z_1)^2} dz_1 \\ &= 2i \prod_{j=1}^n (x_j+i)^{-\alpha_j} \left( z_1 - \frac{x_j-i}{x_j+i} \right)^{-\alpha_j} \frac{dz_1}{(1-z_1)^{2-\alpha_j}} \\ &= C_1 \prod (z_1 - b_j)^{-\alpha_j} dz_1, \end{aligned} \quad (8.3.9)$$

where  $b_j = (a_j - i)/(a_j + i)$ , and the exponent  $2 - \alpha_j$  is zero in the product. Thus, formula (7.1.1) becomes

$$w = C_1 \int_{z_{10}}^{z_1} \prod_{j=1}^n (z_1 - b_j) dz_1 + z_0, \quad (8.3.10)$$

where the points  $b_j$  lie on the unit circle  $|z_1| = 1$ . The lower limit  $z_{10}$  may be chosen as the center of this circle or a point on its circumference. Then there are two cases to consider:

**Case (i)** If  $z_{10} = 0$ , then the integration is carried out along the ray  $z_1 = r e^{i\theta}$ ,  $0 \leq r \leq R$ ,  $\theta = \text{const}$ , and  $b_j = e^{i\phi_j}$ ,  $j = 1, \dots, n$ . Then the mapping function (8.3.10) reduces to

$$\begin{aligned} w &= C_1 e^{i\theta} \int_0^R \prod_{j=1}^n (r e^{i\theta} - e^{i\phi_j})^{-\alpha_j} r dr + z_0 \\ &= C_1 e^{i\theta} \int_0^R \prod_{j=1}^n e^{-i\theta\phi_j} \left( r - e^{i(\phi_j-\theta)} \right)^{-\alpha_j} r dr + z_0 \\ &= C_1 e^{-i\theta} \int_0^R \prod_{j=1}^n \left( r - e^{i(\phi_j-\theta)} \right)^{-\alpha_j} r dr + z_0. \end{aligned} \quad (8.3.11)$$

**Case (ii).** If  $z_{10} = 1$ , we choose the path of integration along the circumference. Thus, for a point  $z_1 = e^{i\theta}$ , the mapping function (8.3.10) becomes

$$\begin{aligned}
 w &= C_1 \int_0^\theta \prod_{j=1}^n (e^{i\theta} - e^{i\phi_j})^{-\alpha_j} i e^{i\theta} d\theta + z_0 \\
 &= i C_1 \int_0^\theta \prod_{j=1}^n e^{-(\sum \phi_j \alpha_j)/2} \left( e^{i(\theta-\phi_j)/2} - e^{-i(\theta-\phi_j)/2} \right)^{-\alpha_j} d\theta + z_0 \\
 &= i(2i)^{-2} C_1 \int_0^\theta \prod_{j=1}^n \left( \sin \frac{\theta - \phi_j}{2} \right)^{-\alpha_j} d\theta + z_0 \\
 &= K \int_0^\theta \prod_{j=1}^n \left( \sin \frac{\theta - \phi_j}{2} \right)^{-\alpha_j} d\theta + z_0, \quad K = i(2i)^2 C_1.
 \end{aligned} \tag{8.3.12}$$

If  $\sin \frac{\theta - \phi_j}{2} < 0$ , then we choose the branch

$$\sin \left( \frac{\theta - \phi_j}{2} \right)^{-\alpha_j} = \exp \left\{ -\alpha_j \log \left| \sin \frac{\theta - \phi_j}{2} \right| - i\pi \alpha_j \right\}.$$

There is no problem for  $\sin \frac{\theta - \phi_j}{2} \geq 0$ . Thus, there exists a constant argument in (8.3.12) in each interval  $\phi_{j-1} < \theta < \phi_j$ ,  $j = 1, \dots, n$ , and the length  $l_j = |z_{j-1}, z_j| = |z_j - z_{j-1}|$  is given by

$$l_j = |K| \int_{\phi_{j-1}}^{\phi_j} \prod_{j=1}^n \left| \sin \frac{\theta - \phi_j}{2} \right|^{-\alpha_j} d\theta, \quad j = 1, \dots, n. \tag{8.3.13}$$

Therefore, the parameter problem for the unit circle is solved by carrying out the integration in (8.3.13) over a finite interval. ■

**8.3.1 Removal of Singularities.** The computation of the improper integrals in the formulas (7.1.1) or (7.1.5) can be easily carried out by removing the singularities in the integrand. There are two analytical methods to do this.

**Method 1.** We will consider the exponent  $-\alpha_1$ ; others can be handled similarly. If  $\alpha_1 > 0$  in (7.1.1), we set

$$K_1 = (x_1 - x_2)^{-\alpha_1} \cdots (x_1 - x_n)^{-\alpha_n}$$

and rewrite the integrand in (7.1.1) as

$$\begin{aligned}
 \int_{z_0}^z &\left[ (\zeta - x_1)^{1-\alpha_1} (\zeta - x_2)^{-\alpha_2} \cdots (\zeta - x_n)^{-\alpha_n} - (\zeta - x_1)^{-\alpha_1} K_1 \right] d\zeta \\
 &+ \frac{K_1}{1 - \alpha_1} \left[ (\zeta - x_1)^{1-\alpha_1} - (z_0 - x_1)^{1-\alpha_1} \right]. \tag{8.3.14}
 \end{aligned}$$

The integrand in (8.3.14) in the neighborhood of  $z = z_1$  is approximately equal to  $(\zeta - x_1)^{-\alpha_1}$ . In fact, it consists of two factors, the first of which goes to  $\infty$  and the second to 0 as  $\zeta \rightarrow x_1$ . Hence, the product is bounded or even goes to zero.

If we use (8.3.14) for part (ii) of Case 8.4, then we get integrals of the form

$$I = \int_{\phi_1}^{\theta} \left( \sin \frac{\theta - \phi_1}{2} \right)^{-\alpha_1} d\theta = 2 \int_0^{(\theta - \phi_1)/2} (\sin t)^{-\alpha_1} dt,$$

where we have set  $\theta = \phi_1 + 2t$ . Now, let  $\phi_1 < \theta < \phi_1 + \pi$ . Then, by setting  $t = \sin^{-1} u$ ,  $u = \sqrt{x}$ , we obtain

$$I = 2 \int_0^{u_0} u^{-\alpha_1} \frac{1}{\sqrt{1-u^2}} du = \int_0^{x_0} x^{-\alpha_1/2} (1-x)^{-1/2} dx, \quad x_0 < 1, \quad (8.3.15)$$

which is the incomplete beta function.

**Method 2.** If  $\alpha_1 > 0$  in (7.1.1), then we remove the singularity at  $\zeta = x_1$  by using the transformation  $z_1 = (z - x_1)^{1-\alpha_1}$ , i.e.,

$$z = x_1 + z_1^{1/(1-\alpha_1)}. \quad (8.3.16)$$

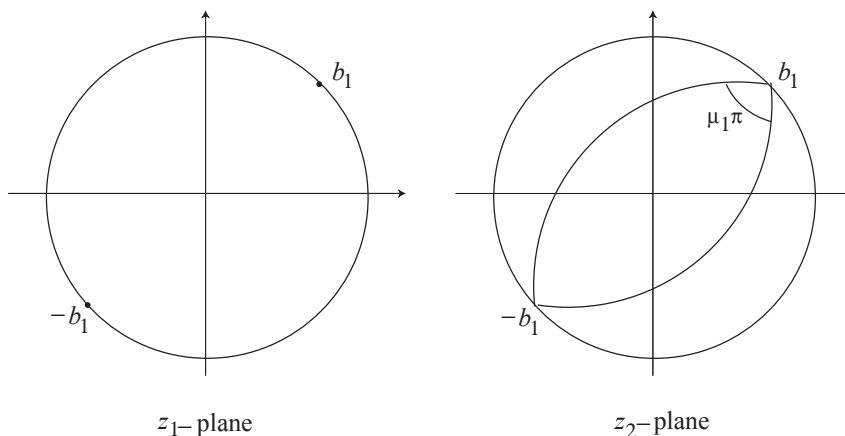


Figure 8.2 A circle onto the region bounded by two circles.

Then the integral (7.1.1) becomes

$$w = z_0 + \frac{C}{\alpha_1} \int_{z_{10}}^{z_1} \left( z_1^{\alpha_1/(1-\alpha_1)} + x_1 - x_2 \right)^{-\alpha_2} \cdots \left( z_1^{-\alpha_1/(1-\alpha_1)} x_1 - x_n \right)^{-\alpha_n} dz_1, \quad (8.3.17)$$

which does not have infinity for  $\alpha_1$ . The transformation (8.3.16) transforms the half-plane  $\Im\{z\} > 0$  onto an angular sector of argument  $(1 - \alpha_j)\pi = \mu_j\pi$ . Note that the mapping (8.3.16) is not suitable for the case of the circle. However, the transformation that maps the circle onto a region bounded by two circles is given by

$$z_1 = b_1 \frac{(b_1 + z_2)^{1/\mu_1} - (b_1 - z_2)^{1/\mu_1}}{(b_1 + z_2)^{1/\mu_1} + (b_1 - z_2)^{1/\mu_1}}, \quad (8.3.18)$$

which is represented in Figure 8.2. ■

**8.3.2 Trefethen's Method.** Let the  $N$  points  $z_k$  (called *prevertices*) be taken in the counterclockwise order around the unit circle and two complex constants  $C$  and  $w_c$ . The Schwarz-Christoffel formula, defined by (7.1.18), is written as

$$w = f(z) = w_c + C \int_0^z \prod_{k=1}^N \left(1 - \frac{z'}{z_k}\right)^{-\alpha_k} dz', \quad (8.3.19)$$

where the variables  $z_1, \dots, z_n, C$ , and  $w_c$  are the accessory parameters of the Schwarz-Christoffel mapping problem. We will first resolve the parameter problem by determining the values of these accessory parameters so that the lengths of the sides of the image polygon comes out right.

**Theorem 8.2.** (Schwarz-Christoffel theorem) *Let  $D$  be a simply connected region in the  $z$ -plane bounded by a polygon  $P$  with vertices  $z_1, \dots, z_n$  and exterior angles  $\alpha_k\pi$ , where  $-1 \leq \alpha_k \leq 1$  if  $z_k$  is finite, and  $1 \leq \alpha_k \leq 3$  if  $z_k = \infty$ . Then there exists an analytic function which maps the unit disk in the complex plane conformally onto  $D$ , and every such function is written in the form (8.3.19).*

For proof, see Henrici [1974, Theorem 5.12e].

Given any polygon, there are not one, but infinitely many such conformal mappings. To determine a unique map, we may fix exactly any three points  $z_k$ , or fix one point  $z_k$  and also fix the complex value  $w_c$ , or fix  $w_c$  and the argument of the derivative  $f'(0)$ . The numerical computation of the Schwarz-Christoffel integral (8.3.19) is carried out as follows.

Gaier [1964] put forth a comprehensive research in numerical conformal mapping. For Schwarz-Christoffel transformation, he worked on determining the accessory parameters  $z_k$  by setting up a constrained nonlinear system of  $N - 3$  equations related to the mapping (8.3.19) and solving it iteratively by Newton's method (Gaier [1964: 171]). This method has been tried by at least three researchers, namely Meyer [1979], Howe [1973], and Vecheslavov and Kokolin [1974].

Following Gaier [1964], three innovations are made to cut computing processing time and cost: (i) use of the Gauss-Jacobi quadrature to evaluate the integral in (8.3.19). In this case it was found that this procedure provided very low accuracy in realistic problems. To overcome this problem, a compound form of Gauss-Jacobi quadrature was developed; the details are given in §8.3.2(c). (ii) The computations may be performed not only for bounded polygons but also for polygons with any number of vertices at infinity. This was made possible by taking the unit disk as the model domain rather than the upper half-plane, and evaluating complex contour integrals within the disk rather than along the boundary. Note that the ability to handle unbounded polygonal regions is important for applications, because the use of conformal mapping lies in reducing an unbounded region into a bounded one. (iii) The treatment of constraints in the nonlinear system is accomplished by a simple change of variables to eliminate these constraints directly. This procedure is efficient and eliminates the need for initial guess of the accessory parameters. The details of these procedures are discussed below.

**(a) Formulation as a nonlinear system (subroutine SCFUN).** The parameters in the map (8.3.19) that need to be fixed at the outset so that the Schwarz-Christoffel transformation may be determined uniquely have the following choices: (i) Fix three of the boundary points  $z_k$ , say,  $z_1 = 1, z_2 = i, z_3 = -i$ . This has the advantage that the resulting nonlinear system has size only  $(N - 3) \times (N - 3)$ , which for a typical problem with  $N = 8$  may lead to a solution in less than one-half the time that a method involving

an  $(N - 1) \times (N - 1)$  system would require. Nevertheless, by normalizing the conditions

$$z_N = 1, \quad w_c \text{ an arbitrary point within } G, \quad (8.3.20)$$

we obtain an  $(N - 1) \times (N - 1)$  system. This choice arises from the idea of numerical scaling, so as to allow the vertices to distribute themselves more evenly around the unit circle than they would otherwise do. Recall that a Möbius transformation will transform a map obtained by any normalization to a different region or a different transformation.

The formulation of the nonlinear system is as follows: Since the final map must satisfy  $N$  complex conditions, we have

$$w_k - w_c = C \int_0^{z_k} \prod_{j=1}^N \left(1 - \frac{z'}{z_j}\right)^{-\alpha_j} dz', \quad 1 \leq k \leq N. \quad (8.3.21)$$

Thus,  $2N$  real conditions must be satisfied; but they are heavily over-determined, because the Schwarz-Christoffel formula (8.3.19) guarantees that the angles will be correct no matter what accessory parameters are chosen. We must reduce the number of operative equations to  $N - 1$ . However, this becomes tricky in the case of unbounded polygons; care is needed in such a situation to have enough information about the polygon  $P$  so that no degree of freedom remains in the computed solution. The procedure is as follows:

STEP 1. We require that every connected component of  $P$  contains at least one vertex  $w_k$ . Even an infinite straight boundary must contain a (degenerate) vertex.

STEP 2. At least one component of  $P$  must actually contain two finite vertices, and  $w_N$  and  $w_1$  will be taken to be two such vertices, thereby eliminating rotational degrees of freedom.

STEP 3. Define

$$C = \frac{w_N - w_c}{\int_0^{z_N} \prod_{j=1}^N \left(1 - \frac{z'}{z_j}\right)^{-\alpha_j} dz'}, \quad (8.3.22)$$

where  $z_N = 1$  is fixed permanently by (8.3.20).

STEP 4. Impose the complex condition (we have two real equations 1, 2)

$$w_1 - w_c = C \int_0^{z_1} \prod_{j=1}^N \left(1 - \frac{z'}{z_j}\right)^{-\alpha_j} dz'. \quad (8.3.23)$$

This amounts to two real equations to be satisfied.

STEP 5. Denote by  $\Gamma_1, \dots, \Gamma_m$  the distinct components of  $P$ , numbered counterclockwise. For each  $l \geq 2$ , impose one more complex condition: If  $z_{k_l}$  is the last vertex of  $\Gamma_l$  in the counterclockwise direction, then we have real equations  $3, 4, \dots, 2m$ :

$$w_{k_l} - w_c = C \int_0^{z_{k_l}} \prod_{j=1}^N \left(1 - \frac{z'}{z_j}\right)^{-\alpha_j} dz'. \quad (8.3.24)$$

STEP 6. Impose  $n - 2m - 1$  conditions of side length. For each pair  $(z_k, z_{k+1})$  beginning at  $k = 1$  and moving counterclockwise, where both vertices are finite, we require that the

following  $2m + 1, \dots, N - 1$  real equations be satisfied:

$$|w_{k+1} - w_k| = \left| C \int_0^{z_{k+1}} \prod_{j=1}^N \left( 1 - \frac{z'}{z_j} \right)^{-\alpha_j} dz' \right|, \quad (8.3.25)$$

until a total of  $N - 1$  conditions have been imposed.

If  $P$  contains at least one vertex at infinity, then every bounded side has already been represented in a condition of the form (8.3.25) except for the side  $(w_N, w_1)$ , which has already been included in (8.3.20) and (8.3.23). If  $P$  is bounded, then at least two sides in counterclockwise order, namely  $(w_{N-2}, w_{N-1})$  and  $(w_{N-1}, w_N)$ , will not be so represented.

In the above presentation we have not defined the contours for the integrals of Eq (8.3.23), (8.3.24) and (8.3.25). Mathematically it does not matter since the integrands are analytic functions. However, numerically it is an important aspect. The procedure discussed here computes these integrals over straight line segments between two endpoints, and it is justified since the unit disk is strictly convex. The contours involved in computing the integrals (8.3.22), (8.3.23), (8.3.24) and (8.3.25) are presented in Figure 8.3 for the simple case with  $N = 10, m = 3$ .

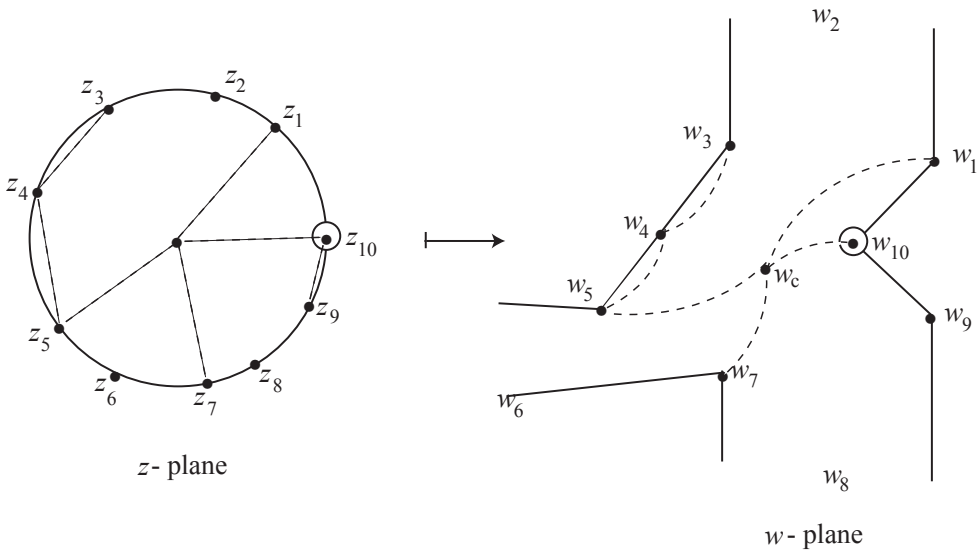


Figure 8.3 Contours of integration within the unit disk.

This figure illustrates, for example, the computation of the following integrations:

- (i) one radical integral along  $(0 \text{ to } z_{10})$  defined by Eq (8.3.22);
  - (ii) one radical integral along  $(0 \text{ to } z_1)$  determines two real equations to fix  $w_1$  (Eq (8.3.23);
  - (iii) two radical integrals along  $(0 \text{ to } z_5)$  and  $(0 \text{ to } z_7)$  determine four real equations to fix  $w_5$  and  $w_7$  (Eq (8.3.24);
  - (iii) three radical integrals along  $(z_3 \text{ to } z_4)$ ,  $(z_4 \text{ to } z_5)$ , and  $(z_9 \text{ to } z_{10})$  determine three real equations to fix  $|w_4 - w_3|$ ,  $|w_5 - w_4|$ , and  $|w_{10} - w_9|$  (Eq (8.3.25);
- thus, the total being  $N - 1 = 9$  equations.

The Fortran subroutine SCFUN, available in Trefethen [1979], was used for the above computations.

**(b) Transformation of an unconstrained system (subroutine YZTRAN).** The nonlinear system of equations (8.3.23)-(8.3.25) apparently involves  $N - 1$  complex unknown points  $z_1, \dots, z_{N-1}$  on the unit circle. In practice we deal not with the points themselves, but their arguments  $\theta_k$  are given by

$$z_k = e^{i\theta_k}, \quad 0 \leq \theta_k \leq 2\pi. \quad (8.3.26)$$

The system now depends on  $N - 1$  real unknowns, and the solution in terms of  $\theta_k$  is fully determined. However, this system of nonlinear equations must be subject to a set of strict inequality constraints

$$0 < \theta_k < \theta_{k+1}, \quad 1 \leq k \leq N - 1. \quad (8.3.27)$$

These constraints arise from the fact that the vertices  $z_k$  must lie in ascending order counterclockwise around the unit circle. To solve this system numerically, we must eliminate these constraints somehow. This is done by transforming Eqs (8.3.23)-(8.3.25) to a system of  $N - 1$  variables  $y_1, \dots, y_{N-1}$ , defined by

$$y_k = \log \frac{\theta_k - \theta_{k-1}}{\theta_{k+1} - \theta_k}, \quad 1 \leq k \leq N - 1, \quad (8.3.28)$$

where  $\theta_0$  and  $\theta_N$ , which are the two different names for the argument of  $z_N = 1$ , are taken as 0 and  $2\pi$ , respectively. While solving the nonlinear system (8.3.23)-(8.3.25), we begin, at each iteration, by computing a set of angles  $\{\theta_k\}$  and then vertices  $\{z_k\}$  from the current set of  $\{y_k\}$ . However, since the equations (8.3.28) are coupled, it is easy to do, though not immediate. This procedure reduces the problem to one of solving an unconstrained nonlinear system of equations in  $N - 1$  real variables.

**(c) Integration by compound Gauss-Jacobi quadrature (subroutine ZQUAD).**

The main aim for computing the parameter problem is the numerical evaluation of the Schwarz-Christoffel integral (8.3.19) along some path of integration. Typically, one or both endpoints of this path are prevertices  $z_k$  on the unit circle, and in this case a singularity of the form  $(1 - z/z_k)^{-\alpha_k}$  is always present in the integrand at one or both endpoints. Naturally, Gauss-Jacobi quadrature (see Appendix G) computes such integrals quickly. A

Gauss-Jacobi quadrature formula is a sum  $\sum_{i=1}^{\text{NPTS}} w_i f(x_i)$ , where the weights  $w_i$  and nodes

$x_i$  are chosen in such a way that the formula computes the integral  $\int_{-1}^{+1} f(x)(1-x)^a(1+x)^\beta dx$  exactly for a polynomial  $f(x)$  of as high a degree as possible, and NPTS denotes the number of quadrature nodes per half-interval. The required weights and nodes are computed numerically, using a Fortran program GAUSSQ by Golub and Welsch [1969].

Using Gauss-Jacobi quadrature, good results were obtained for many polygons with a small number of vertices. However, in general, this method of integration became inaccurate for  $N = 12$  and NPTS = 16, where it produced inaccuracy of about  $10^{-2}$ . The results became much worse in cases of troublesome polygons. To illustrate this situation, consider a problem of Figure 8.4, where we want to compute the integral (8.3.19) along the segment from  $z_k$  to some point  $p$ . In the parameter problem  $p$  might be 0 or  $z_{k-1}$ ; or it might be any point in the unit disk. A direct application will then involve sampling the integrand at only NPTS nodes between  $z_k$  and  $p$ . If the singularity  $z_{k+1}$  is so close to the path of integration that the distance  $\varepsilon = |z_{k+1} - z_k|$  is comparable to the distance between nodes,



then Gauss-Jacobi formula will give an inaccurate result. Also, it is found that in Schwarz-Christoffel problems the correct spacing of prevertices  $z_k$  around the unit circle is typically very irregular; examples are given in Chapter 7.

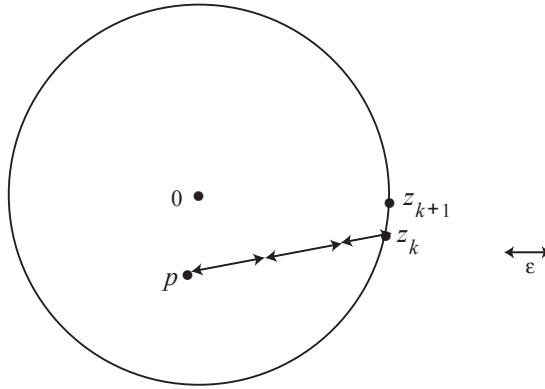


Figure 8.4 Compound Gauss-Jacobi quadrature.

This figure shows the division of an interval of integration into subintervals to maintain the desired resolution. To maintain high accuracy with much speed, a compound Gauss-Jacobi quadrature, provided in Davis and Rabinowitz [1975: 56], is adopted, with the quadrature principle

No singularity  $z_k$  shall lie closer to an interval of integration than half the length of that interval.

To achieve this principle, the quadrature subroutine ZQUAD must be able to divide an interval of integration into shorter subintervals as needed, working from the endpoints inward. On the short subinterval adjacent to the endpoint, Gauss-Jacobi quadrature is applied; on longer interval(s) away from the endpoint, pure Gaussian quadrature is applied. The effect of this procedure is that the number of integrand evaluations required to achieve a given accuracy is reduced from  $O(1/\varepsilon)$  to  $O(\log_2(1/\varepsilon))$ . For example, in the 12-vertex example of Figure 8.3, the switch to compound Gauss-Jacobi integration decreases the error from  $10^{-2}$  to  $2 \cdot 10^{-7}$ .

There is only one case where the above-described integration by compound Gauss-Jacobi quadrature fails. This case involves an integration interval with one point very close to some prevertex  $z_k$  corresponding to a vertex  $w = \infty$ . This integral cannot be computed, considering an interval that begins at  $z_k$ , because the integral then becomes infinite. However, a proper procedure in this case is integration by parts, which can reduce the singular integrand to one that is not infinite. Depending on the angle  $\alpha_k$ , one would need one to three applications of integration by parts. However, this procedure has not been implemented in the subroutine ZQUAD.

**(d) Solution of system by packaged solver (subroutine SCSOLV).** After the above discussions, the unconstrained nonlinear system is ready to be computed. A library subroutine NSOIA (by Powell [1968]) is employed; it uses a steepest descent search in early iterations if necessary followed by a variant of Newton's method later on.

Fortran programs of all subroutines are available in Trefethen [1979: 48-55].

## 8.4 Minimum Area Problem

Gaier's variational method is used to solve two extremal problems in the theory of conformal mapping. The first deals with the conformal mapping of a simply connected region onto a disk, and the second with that of the boundary of the region onto the circumference of the disk. Both problems use the Ritz method for approximating the minimal mapping function by polynomials. This mapping function in the first problem is represented in terms of the Bergman kernel function, and in the second problem in terms of the Szegő kernel.

An extremal property in the conformal mapping of a simply connected region  $D$  onto a disk is connected with the minimum area problem. This problem, which we will denote as Problem I, is known as the Bieberbach minimizing principle. The mapping function possesses an extremal property which provides a method to compute an approximate solution for the map. Let  $\mathcal{K}^1(D)$  denote the class of all functions  $f \in L^2(D)$  with  $f(a) = 1$ , where  $a \in D$ . Similarly, let  $\mathcal{K}^0(D)$  denote the class of all functions  $f \in L^2(D)$  such that  $f(a) = 0$ . Note that the classes  $\mathcal{K}^1(D)$  and  $\mathcal{K}^0(D)$  represent, respectively, a closed convex subset and a closed subspace of  $L^2(D)$ . Let the function

$$w = f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n, \quad |z - a| < R, \quad (8.4.1)$$

which is regular in  $D$ , map  $D$  onto the disk  $B(0, R)$  in the  $w$ -plane. Without loss of generality, we will sometimes take the point  $a$  as the origin.

We will designate the *minimum area problem* as Problem I, defined as follows:

**Problem I:** In the class  $\mathcal{K}^1$ , minimize the integral

$$I = \iint_D |f'(z)|^2 dS_z. \quad (8.4.2)$$

The Riemann mapping theorem (Theorem 2.15) guarantees the existence and uniqueness of the solution of this extremal problem.

**Theorem 8.3.** *Problem I (minimum area problem) has a unique solution  $f_0(z) = f'(z)$ . The minimum is  $\pi R^2$ .*

PROOF. If  $f_0(a) = 0$  and  $f'_0(a) = 1$ , then in view of (2.6.25)

$$\begin{aligned} \text{area}(D) &= \iint_D |f'(z)|^2 dS_z = \int_0^R \int_0^{2\pi} |f'(r e^{i\theta})|^2 r dr d\theta \\ &= \int_0^R \sum_{n=0}^{\infty} |a_n|^2 n^2 2\pi r^{2n-1} dr \\ &= \pi R^2 |a_1|^2 + \pi \sum_{n=2}^{\infty} n |a_n|^2 R^{2n}. \end{aligned} \quad (8.4.3)$$

The above result implies that in the problem of mapping by the function (8.4.1), which is regular in  $B(0, R)$  and is such that  $f'(a) = a_1$ , the area of the mapped region  $D$  is always greater than  $\pi R^2 |a_1|^2$ . It is exactly equal to this value if the map  $f(z)$  is linear, i.e., if  $w = a_0 + a_1 z$ . A particular case is when  $a_1 = 1$ . Then the mapping function is  $w = a_0 + z$ . If this linear transformation is excluded, then the mapping function can be normalized by the conditions  $f(a) = 0$  and  $f'(a) = 1$ , by considering the function  $f(z)/a_1$ . In either case the minimum area of  $D$  is  $\pi R^2$ . ■

Before we solve Problem I, we will examine the minimal function  $f_0(z)$  closely.

**Theorem 8.4.** *The function  $f_0(z)$  is orthogonal to every function  $g \in L^2(D)$  with  $g(a) = 0$ , i.e.,*

$$\langle f_0, g \rangle = \iint_D f_0(z) \overline{g(z)} dS_z = 0. \quad (8.4.4)$$

PROOF. For every  $\varepsilon > 0$  and  $0 \leq \theta \leq 2\pi$ , the function  $f_0(z) + \varepsilon g(z)$  belongs to the class  $\mathcal{K}^1$ . Then

$$\begin{aligned} \iint_D |f_0(z) + \varepsilon g(z)|^2 dS_z &\leq \iint_D |f_0(z)|^2 dS_z + 2\varepsilon \Re \left\{ \iint_D f_0(z) \overline{g(z)} dS_z \right\} + \varepsilon^2 \iint_D |g(z)|^2 dS_z, \end{aligned}$$

which implies that

$$\Re \left\{ \iint_D f_0(z) \overline{g(z)} dS_z \right\} + \frac{\varepsilon}{2} \iint_D |g(z)|^2 dS_z \geq 0.$$

If (8.4.4) were false, then the above expression would be negative for sufficiently small  $\varepsilon > 0$ . ■

**8.4.1 Bergman Kernel.** If we take  $g(z) = f_0(z) - f_0(a)$ , then from (8.4.4) for  $f \in \mathcal{K}^1(D)$  we have

$$\iint_D f_0(z) \overline{f(z)} dS_z = \overline{f(a)} \iint_D f_0(z) dS_z,$$

and if  $f = f_0$ , then

$$\|f_0\|^2 = \iint_D |f_0(z)|^2 dS_z = \overline{f_0(a)} \iint_D f_0(z) dS_z.$$

If we introduce the Bergman kernel

$$K(z, a) = \frac{f_0(z)}{\|f_0\|^2}, \quad (8.4.5)$$

where  $f_0(z)$  minimizes Problem I, then for every  $f \in \mathcal{K}^1(D)$

$$\iint_D \overline{K(z, a)} f(z) dS_z = f(a). \quad (8.4.6)$$

Hence, every function  $f \in L^2(D)$  is the eigenfunction of the integral equation (8.4.6) with eigenvalue  $\lambda = f_0(a) = 1$ . Then the minimal function  $f_0(z)$  is, in view of (8.4.5), given by

$$f_0(z) = \frac{K(z, a)}{K(a, a)}, \quad (8.4.7)$$

where

$$K(a, a) = \iint_D |K(z, a)|^2 dS_z = \frac{1}{\|f_0\|^2}. \quad (8.4.8)$$

Note that we cannot find  $f_0(z)$  directly. We can find  $f'(z)$  since it appears in the integrand in (8.4.2). Then the mapping function  $f$  is related to the Bergman kernel of  $D$  by

$$f(z) = \frac{\iint_D K(z, a) dS_z}{K(a, a)}. \quad (8.4.9)$$

## 8.5 Numerical Methods for Minimum Area Problem

We will study the Ritz method (RM) and the Bergman kernel method (BKM) for the minimum area problem (Problem I).

**8.5.1 Ritz Method.** The Ritz method is used to find the solution of the above extremal problem approximately in the form of a polynomial. Consider an arbitrary system of linearly independent functions  $u_0(z), u_1(z), \dots$ , which are regular in  $D$  and are such that  $\iint_D |u_k(z)|^2 dx dy < +\infty$  for  $k = 0, 1, \dots$ . We assume that one of these functions, say  $u_0(z)$ , is such that  $u_0(a) \neq 0$ . Without loss of generality, we take  $a = 0$ . Let  $\{\phi_n(z)\}$  be a complete set of  $L^2(D)$ , and denote by  $\mathcal{K}_n^0$  and  $\mathcal{K}_n^1$  the  $n$ -dimensional counterparts of  $\mathcal{K}^0$  and  $\mathcal{K}^1$ , respectively, i.e., if

$$\phi_n(z) = \sum_{k=0}^n c_k u_k(z), \quad (8.5.1)$$

then  $\phi_n \in \mathcal{K}_n^0$  if  $\phi(0) = 0$ , and  $\phi_n \in \mathcal{K}_n^1$  if  $\phi_n(0) = 1$ , where  $c_k = \alpha_k + i\beta_k$ ,  $c_0 \neq 0$ , are complex numbers. Now we will consider the integral

$$I(\phi_n) = \iint_D |\phi_n(z)|^2 dS_z, \quad (8.5.2)$$

which is the same as (8.4.1) except that the integrand in (8.5.2) is  $\phi_n(z)$  instead of  $f'(z)$ .

**Problem I<sub>n</sub>:** In the class  $\mathcal{K}_n^1$ , minimize the integral  $I(\phi_n)$  defined by (8.5.2).

Now we will discuss the existence and uniqueness of the minimal polynomial  $\phi_n(z)$ , determine  $\phi_n(z)$ , and approximate  $f_0(z)$  by the minimal polynomial  $\phi_n(z)$ . The numerical value of the integral (8.5.2) is equal to the area of the image of the region  $D$ . Then the problem reduces to a choice of the coefficients  $c_k$  so that this value is a minimum among the values of the same integral for any other linear combination  $\psi_n$  of the functions  $u_k(z)$ ,  $k = 0, 1, \dots, n$ , subject to the condition  $\psi_n(0) = 1$ . Suppose that

$$\psi_n(z) = \phi_n(z) + \varepsilon \gamma_n(z), \quad (8.5.3)$$

where  $\varepsilon$  is a complex number and  $\gamma_n(z)$  is a linear combination of  $u_k(z)$ ,  $k = 0, 1, \dots, n$ . The requirement that  $\psi_n(0) = 1$  for all  $\varepsilon$  implies that  $\gamma_n(0) = 0$ . Now

$$\begin{aligned} I(\psi_n) &= \iint_D |\phi_n|^2 dx dy + \bar{\varepsilon} \iint_D \phi_n \bar{\gamma}_n dx dy \\ &\quad + \varepsilon \iint_D \bar{\phi}_n \phi_n dx dy + |\varepsilon|^2 \iint_D |\gamma_n|^2 dx dy. \end{aligned} \quad (8.5.4)$$

The sign of the difference  $I(\psi_n) - I(\phi_n)$  for small  $\varepsilon$  will depend on the linear terms in  $\varepsilon$  and  $\bar{\varepsilon}$  because the last term in (8.5.4) is of order  $O(\varepsilon^2)$ . Thus,  $I(\psi_n) - I(\phi_n) \geq 0$  iff the following orthogonality relations hold:

$$\iint_D \phi_n \bar{\gamma}_n dx dy = 0 \quad \text{and} \quad \iint_D \bar{\phi}_n \gamma_n dx dy = 0 \quad (8.5.5)$$

for any linear combination  $\gamma_n$  that satisfies the condition  $\gamma_n(0) = 0$ . Otherwise, we can always choose an  $\varepsilon$  which will make  $I(\psi_n) < I(\phi_n)$ , and this will contradict the minimal properties of  $\phi_n(z)$ . Note that the integrands in (8.5.5) are complex conjugates of each other, so we can use either as needed.

Conversely, if  $\phi_n(z)$  satisfies the orthogonality relations (8.5.5), then  $\phi_n(z)$  imparts the integral  $I(\phi_n)$  its minimum value among the values imparted by all linear combinations of  $\psi_n(z)$  with  $\psi_n(0) = 1$ . Thus, from (8.5.4) and (8.5.5) we get

$$I(\psi_n) - I(\phi_n) = |\varepsilon|^2 \iint_D |\gamma_n|^2 dx dy. \quad (8.5.6)$$

Hence, the polynomial  $\phi_n(z)$  will be unique if it exists, since the integral on the right side of (8.5.6) is equal to zero only when  $\gamma_n = 0$ , i.e., when  $\psi_n = \phi_n$ . Thus, the orthogonality relations (8.5.5) constitute necessary and sufficient conditions for  $\phi_n(z)$  to be the minimal polynomial.

We will rewrite the conditions (8.5.5) in a different but equivalent form. Let

$$v_k(z) = u_k(z) - \frac{u_k(0)}{u_0(0)} u_0(z), \quad k = 1, \dots, n. \quad (8.5.7)$$

Then each of the functions  $v_k(z)$  satisfies the requirements imposed on  $\gamma_k(z)$ , and the conditions (8.5.5) become

$$\sum_{j=0}^n A_{kj} c_j = 0, \quad (8.5.8)$$

where

$$A_{kj} = \iint_D \bar{u}_j v_k dx dy = \iint_D u_j \bar{v}_k dx dy, \quad k = 1, \dots, n. \quad (8.5.9)$$

Also, since

$$\sum_{k=0}^n u_k(0) c_k = 1, \quad (8.5.10)$$

Eqs (8.5.8) and (8.5.10) provide us with a system of  $(n+1)$  equations to determine the numbers  $c_0, c_1, \dots, c_n$  uniquely.

Note that the functions  $u_k(z)$ , though linearly independent, are still undetermined and are not subject to limitation. However, it becomes very easy to determine the integrals in (8.5.9) if all  $u_k(z)$  are suitably chosen beforehand such that  $u_0(0) = 1$ , and  $u_k(0) = 0$  for  $k = 1, \dots, n$ . Then  $v_k(z) = u_k(z)$ , and the integral

$$\iint_D u_j \bar{v}_k dx dy = \iint_D u_j \bar{u}_k dx dy.$$

Also, Eq (8.5.10) degenerates to  $c_0 = 1$ , which reduces the number of unknowns by one. Moreover, if the system of functions  $u_0(z), u_1(z), \dots$  is orthogonal in  $D$ , i.e., if

$$\iint_D u_j \bar{u}_k dx dy = 0 \quad \text{for } k \neq j,$$

then

$$\iint_D u_j \bar{v}_k dx dy = 0 \quad \text{for } j \neq 0 \text{ and } k \neq j,$$

and

$$\iint_D u_0 \bar{v}_k dx dy = -\frac{\overline{u_k(0)}}{u_0(0)} \iint_D u_0 \bar{u}_0 dx dy,$$

$$\iint_D u_k \bar{v}_k dx dy = \iint_D u_k \bar{u}_k dx dy.$$

Then Eqs (8.5.8)-(8.5.10) simplify to

$$c_k \iint_D |u_k|^2 dx dy - c_0 \frac{\overline{u_k(0)}}{u_0(0)} \iint_D |u_0|^2 dx dy = 0. \quad (8.5.11)$$

Thus, we have proved

**Theorem 8.5.** *The problem  $I_n$  has a unique solution. The coefficients of the minimal polynomial  $\phi_n(z)$ , defined by (8.5.1), are determined by the system of linear equations (8.5.8)-(8.5.9). The coefficients  $A_{kj}$ , defined by (8.5.9), are hermitian.*

Let the system  $\{u_k\}_{k=0}^n$  be taken as the complete set  $\{1, z, z^2, \dots, z^n\}$ . Then for the minimal polynomial

$$\phi_n(z) = 1 + c_1 z + c_2 z^2 + \dots + c_n z^n, \quad (8.5.12)$$

the system of equations (8.5.8)-(8.5.9), which determine the  $n$  coefficients  $c_k$ , becomes

$$\begin{bmatrix} A_{10} & A_{11} & A_{12} & \cdots & A_{1n} \\ A_{20} & A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ A_{n0} & A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix} \begin{Bmatrix} 1 \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{Bmatrix} = \{0\}, \quad (8.5.13)$$

where the coefficients  $A_{kj}$  are given by

$$A_{kj} = \iint_D z^k \overline{z^j} dS_z, \quad A_{kj} = \bar{A}_{jk}. \quad (8.5.14)$$

Note that if  $a \neq 0$ , then the coefficients  $A_{kj}$  are determined from

$$A_{kj} = \iint_D (z - a)^k \overline{(z - a)^j} dS_z, \quad A_{kj} = \bar{A}_{jk}. \quad (8.5.15)$$

The computation of the coefficients  $A_{kj}$  depends on the region  $D$ , although it may sometimes present difficulties.

**Example 8.1.** Let the region  $D$  be starlike with respect to a point  $a \neq 0 \in D$ , i.e., every ray emanating from the point  $a$  intersects the boundary in only one point. Let the equation of the boundary be  $r = r(\theta)$ . Using the polar coordinates  $z - a = r e^{i\theta}$ , we get

$$\begin{aligned} A_{kj} &= \int_0^{2\pi} \int_0^{r(\theta)} r^{j+k} e^{i(j-k)\theta} r dr d\theta = \frac{1}{j+k+2} \int_0^{2\pi} r^{j+k+2}(\theta) e^{i(j-k)\theta} d\theta \\ &= \frac{1}{j+k+2} \int_0^{2\pi} r^{j+k+2}(\theta) \cos(j-k)\theta d\theta \\ &\quad + \frac{i}{j+k+2} \int_0^{2\pi} r^{j+k+2}(\theta) \sin(j-k)\theta d\theta. \end{aligned} \quad (8.5.16)$$

Note that  $\Re \{A_{kj}\}$  and  $\Im \{A_{kj}\}$  differ from the coefficients of the Fourier series for  $r^{j+k+2}(\theta)$  by a factor  $\frac{\pi}{j+k+2}$ , and hence, they can be easily computed. ■

**DEFINITION 8.5.1.** The system of polynomials  $\{\phi_n\}$  is said to be *complete* in the Hilbert space  $L^2(D)$  if for every function  $f \in L^2(D)$  and every  $\varepsilon > 0$  there exists a polynomial  $\phi_n$  such that  $\|f - \phi_n\| < \varepsilon$ .

Now the question arises, under what additional assumptions on  $D$  the polynomials  $\phi_n(z)$  form a complete system in  $L^2(D)$ . Naturally,  $\|f_0 - \phi_n\| \searrow 0$  must hold, and thus also  $\phi_n(z) \rightarrow f_0(z)$  as  $n \rightarrow \infty$  in any closed subset  $\bar{G} \subset D$ .

**Theorem 8.6.** Let the polynomial  $p(z)$  belong to the class  $\mathcal{K}_n^1$ . Then

$$\|f_0 - p\|^2 = \iint_D |f_0(z) - p(z)|^2 dS_z \quad (8.5.17)$$

is minimal only if  $p(z) = \phi_n(z)$ .

**PROOF.** In view of (8.4.4)

$$\langle f_0, p \rangle = \langle f_0, p - f_0 \rangle + \langle f_0, f_0 \rangle = \langle f_0, f_0 \rangle,$$

thus,

$$\langle f_0 - p, f_0 - p \rangle = \langle f_0, f_0 \rangle - 2\langle f_0, f_0 \rangle + \langle p, p \rangle = \|p\|^2 - \|f_0\|^2,$$

and the result follows since  $\|p\|^2$  is minimal if  $p = \phi_n$ . ■

Hence, among all  $p \in \mathcal{K}_n^1$  the polynomial  $\phi_n$  yields the minimum norm  $\|f_0 - p\|$ , and since  $\mathcal{K}_{n+1}^1 \supset \mathcal{K}_n^1$ , then  $\|f_0 - p\| \searrow 0$  only if the system of polynomial  $\{\phi_n\}$  is complete in the space  $L^2(D)$ . Moreover, in Problem  $I_n$  there exists a polynomial  $p = \phi_n$  which satisfies the additional condition  $p(a) = f(a)$ . Also note that the Bieberbach polynomial  $\pi_n(z)$  is defined by

$$\pi_n(z) = \int_a^z \phi_{n-1}(\zeta) d\zeta \rightarrow f(z) \quad \text{as } n \rightarrow \infty \text{ in } \bar{G} \subset D. \quad (8.5.18)$$

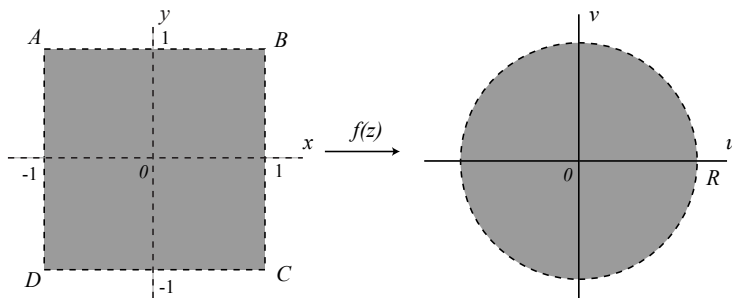


Figure 8.5 Square onto a circle.

**Example 8.2.** Determine the minimal polynomial  $\phi_n(z)$  and the approximate mapping function  $f(z)$  that maps the square region  $D = \{x, y : -1 < x, y < 1\}$  conformally onto  $|w| < 1$  (Figure 8.5). From `cs422.nb` (see Notes at the end of this chapter), the minimum polynomial is given by

$$\phi_8(z) = 1 + \frac{97402305}{266254834} z^4 + \frac{68765697}{2130038672} z^8,$$

which yields the approximate mapping function

$$\begin{aligned} f(z) &\approx \int_0^z \phi_8(t) dt \\ &= z + \frac{19480461}{266254834} z^5 + \frac{7640633}{2130038672} z^9 \\ &\approx z + 0.0731647 z^5 + 0.00358709 z^9. \blacksquare \end{aligned}$$

**8.5.2 Bergman Kernel Method.** In this method the mapping function  $f(z)$  is determined approximately from (8.4.19) by first approximating the kernel  $K(z, 0)$  by a finite Fourier sum. Let  $\{\phi_j^*(z)\}$  denote a complete orthonormal set of  $L^2(D)$ . Consider the Fourier series expansion of  $K(z, 0)$ . Then, in view of (8.4.6),

$$\langle K, \phi_j^* \rangle = \iint_D K(z, 0) \overline{\phi_j^*(z)} dS_z = \overline{\phi_j^*(0)}. \quad (8.5.19)$$

Thus, the kernel has a series expansion

$$K(z, 0) = \sum_{j=1}^{\infty} \overline{\phi_j^*(0)} \phi_j^*(z), \quad (8.5.20)$$

which converges in the mean of  $L^2(D)$ , i.e., the series (8.5.20) converges almost uniformly in  $D$ .

Hence, if we have a complete set  $\{\phi_j(z)\}$  of  $L^2(D)$ , then by using (8.4.19) and (8.5.20) we obtain an approximate mapping function  $f(z)$  as follows:

(i) Orthonormalize the set  $\{\phi_j(z)\}_{j=1}^n$  by using the Gram-Schmidt process which yields the



set of orthonormal functions  $\{\phi_j^*(z)\}_{j=1}^n$ . Note that the Gram-Schmidt process requires evaluating  $\langle \phi_i, \phi_j \rangle$ , which, in view of Green's formula (2.3.11), is given by

$$\langle \phi_i, \phi_j \rangle = \iint_D \phi_i(z) \overline{\phi_j(z)} dS_z = \frac{1}{2i} \int_{\Gamma} \phi_i(z) \overline{\psi_j(z)} dz, \quad (8.5.21)$$

where  $\psi_j'(z) = \phi_j(z)$ . Then the integrals in (8.5.21) are computed by Gaussian quadrature. (ii) Truncate the series (8.5.20) after  $n$  terms to obtain the approximation  $K_n(z, 0)$  of  $K(z, 0)$  as

$$K_n(z, 0) = \sum_{j=1}^n \overline{\phi_j^*(0)} \phi_j^*(z). \quad (8.5.22)$$

(iii) Use (8.4.19) to obtain the approximate mapping function  $f_n(z)$  as

$$f_n(z) = \frac{\int_0^z K_n(z, 0) dz}{K_n(0, 0)}. \quad (8.5.23)$$

(iv) The approximate radius  $R_n$  of the disk  $|w| < R$  is given by

$$R_n = \frac{1}{\sqrt{\pi K_n(0, 0)}}, \quad (8.5.24)$$

since  $\|f_0\|^2 = \pi R^2$  (because  $f_0 \in \mathcal{K}^1$ ).

(v) Thus, from (8.5.23) and (8.5.24) the approximation of the mapping function  $F(z)$  that maps  $D$  conformally onto the unit disk  $|w| < 1$  is given by

$$F_n(z) = \sqrt{\frac{\pi}{K_n(0, 0)}} \int_0^z K_n(z, 0) dz. \quad (8.5.25)$$

The maximum error estimate for  $|F_n(z)|$  is given by

$$E_n = \max_j |e_n(z_j)|,$$

where  $z_j \in \Gamma$  are the test points on the boundary and  $e_n(z) = 1 - |F_n(z)|$ . During the computation process the number  $n$  of the basis function is increased by one each time and this process is terminated when the inequality  $E_{n+1} < E_n$  no longer holds. Then such a number  $n$  is taken as the 'optimum number' for the basis functions.

Note that in both RM and BKM we have obtained approximations of the form

$$f_n(z) = \sum_{j=1}^n a_j u_j(z),$$

where  $u_j'(z) = \phi_j(z)$ . In both methods the set of monomials  $z^{j-1}$ ,  $j = 1, 2, \dots$ , which is a complete set in  $L^2(D \cup \Gamma)$  is the best choice of basis functions in computation. Then this basis gives the polynomials  $\phi_j(z)$  defined in (8.5.13).

**Map 8.2.** The function  $w = f(z)$  that maps the unit disk  $U$  onto itself such that the point  $z_0 \in U$  goes into the origin of the  $w$ -plane is given by

$$w = f(z) = \frac{z - z_0}{z - 1/\bar{z}_0}.$$

Thus, both  $f(z)$  and the associated Bergman kernel function  $K(z, z_0)$  have a pole at  $z = 1/\bar{z}_0$ . Since the polynomials  $\phi_j^*(z) = \sqrt{j/\pi} z^{j-1}$ ,  $j = 1, 2, \dots$ , form a complete orthonormal basis set of  $U$ , the kernel  $K(z, z_0)$  can, in view of (8.5.20), be represented by the polynomial series

$$K(z, z_0) = \frac{1}{\pi} \sum_{j=1}^{\infty} j (\bar{z}_0 z)^{j-1},$$

which converges rapidly when  $|z_0|$  is small, but the convergence becomes considerably slower the faster  $|z_0| \rightarrow 1$ , i.e., the closer the pole  $1/\bar{z}_0$  gets to the boundary of  $U$ . ■

## 8.6 Minimum Boundary Problem

An analogous minimum problem in the conformal mapping of a region  $D$  onto  $|w| < R$  leads to another characterization of the mapping function  $f(z)$  by considering the line integral

$$I = \int_{\Gamma} |f(z)|^2 ds, \quad (8.6.1)$$

where  $\Gamma$  is the boundary of  $D$ . This problem, studied by Julia [1931], is known as the minimum boundary problem which we will call Problem II. Let  $\Gamma$  be a rectifiable Jordan curve, and let  $\mathcal{L}^1(\Gamma)$  denote the class of all functions  $f \in L^2(\Gamma)$  with  $f(a) = 1$ , where  $a \in D$  can be taken as the origin.

**Problem II:** In the class  $\mathcal{L}^1(\Gamma)$  minimize the integral (8.6.1).

**Theorem 8.7.** *Problem II has a unique solution  $f_0(z)$ , and it is  $f_0(z) = \sqrt{f'(z)}$ . The minimum is  $2\pi R$ .*

PROOF. For every function  $F \in L^2(\Gamma)$

$$\begin{aligned} \int_{\Gamma} |F(z)|^2 ds &= \lim_{r \rightarrow R} \int_{\Gamma_r} |F(z)|^2 ds = \lim_{r \rightarrow R} \int_{|w|=r} |F(g(w)) \sqrt{g'(w)}|^2 |dw| \\ &= \lim_{r \rightarrow R} r \int_0^{2\pi} |h(re^{i\theta})|^2 d\theta, \end{aligned}$$

where  $h(w) = F(g(w)) \sqrt{g'(w)}$ ,  $h(0) = 1$ , and  $z = g(w)$  (see (2.3.8)). If  $h(w) = \sum_{n=0}^{\infty} a_n w^n$ ,  $a_0 = 1$ , then

$$\int_{\Gamma} |F(z)|^2 ds = 2\pi \sum_{n=0}^{\infty} |a_n|^2 R^{2n+1} \geq 2\pi R,$$

where the equality holds only for  $a_n = 0$ ,  $n > 0$ , which yields  $F(z) = \sqrt{f'(z)}$  for  $h(w) = 1$ . ■

This theorem implies that in the class of all conformal mappings  $w = \phi(z)$  of the region  $D$  with  $\phi(a) = 0$ ,  $\phi'(a) = 1$ , the integral  $\int_{\Gamma} |\phi'(z)| ds$  is minimum only when  $\phi(z) = f(z)$ . This is known as the *principle of minimizing the image boundary*.

The conformal map  $f$  of  $D$  onto  $|w| < R$ , normalized by  $f(a) = 0$ ,  $f'(a) = 1$ , is given by

$$f(z) = \int_a^z [f_0(\zeta)]^2 d\zeta.$$

The theory for Problem II is developed exactly on the same lines as in §8.4 and §8.5. Thus, as in (8.4.4), we have

**Theorem 8.8.** *The function  $f_0(z)$  is orthogonal to every function  $g \in L^2(\Gamma)$  with  $g(a) = 0$ , i.e.,*

$$\langle f_0, g \rangle = \int_{\Gamma} f_0(z) \overline{g(z)} ds = 0. \quad (8.6.2)$$

For the minimal function  $f_0(z)$  of Problem II we introduce the Szegő kernel function

$$S(z, a) = \frac{f_0(z)}{\int_{\Gamma} |f_0(z)|^2 ds}, \quad (8.6.3)$$

with the properties

$$S(a, a) = \int_{\Gamma} |S(z, a)|^2 ds, \quad \text{and} \quad f_0(z) = \frac{S(z, a)}{S(a, a)}. \quad (8.6.4)$$

For any  $f \in L^2(\Gamma)$ , we apply (8.6.2) to  $f(z) - f(a)$  and get

$$\int_{\Gamma} f_0(z) \overline{f(z)} ds = \overline{f(a)} \int_{\Gamma} f_0(z) ds,$$

and if  $f = f_0$ , then, since  $h(w) = 1$  implies  $f(a) = 1 = \overline{f(a)}$ , we have

$$\int_{\Gamma} |f_0(z)|^2 ds = \int_{\Gamma} f_0(z) ds.$$

Thus, analogous to (8.4.6) we have: For every function  $f \in L^2(\Gamma)$

$$\int_{\Gamma} \overline{S(z, a)} f(z) ds = f(a). \quad (8.6.5)$$

Theorems 8.3 and 8.7 together with the definitions (8.4.5) and (8.6.3) yield

$$K(z, a) = 4\pi [S(z, a)]^2. \quad (8.6.6)$$

Thus,  $S(z, a)$  can be evaluated by this method.

**8.6.1 Ritz Method for Problem II.** Let  $\mathcal{L}_n^1$  denote the class of all polynomials  $p(z)$  of degree  $\leq n$  with  $p(a) = 1$ .

**Problem II<sub>n</sub>:** In the class  $\mathcal{L}_n^1$  minimize the line integral  $\int_{\Gamma} |p(z)|^2 ds$ .

We will discuss the existence and uniqueness of the minimal polynomial  $\phi_n(z)$ , determine  $\phi_n(z)$ , and approximate  $f_0(z)$  by the minimal polynomial  $\phi_n(z)$  and  $f(z)$  by the integral  $\int_a^z [\phi_n(\zeta)]^2 d\zeta$ , respectively. As in §8.5, it can be shown that a minimal polynomial  $\phi_n(z)$  exists for Problem II, that it is unique, and that it is characterized by

$$\langle \phi_n, g \rangle = 0 \quad (8.6.7)$$

for every polynomial  $g(z)$  of degree  $\leq n$  with  $g(a) = 0$ . The proof is analogous to that of (8.4.4).

In view of (8.6.7), the coefficients of the minimal polynomial

$$\phi_n(z) = 1 + a_1(z - a) + \cdots + a_n(z - a)^n \quad (8.6.8)$$

are determined by

$$\int_{\Gamma} \left( \sum_{k=0}^n a_k (z - a)^k \right) \overline{(z - a)^j} ds = 0, \quad j = 1, 2, \dots, n, \quad (8.6.9)$$

and  $a_0 = 1$ . If we set

$$B_{kj} = \int_{\Gamma} (z - a)^k \overline{(z - a)^j} ds, \quad k, j = 0, 1, \dots, \quad (8.6.10)$$

then the coefficients of the minimal polynomial  $\phi_n(z)$  defined by (8.6.8) are determined from the (consistent) system of equations

$$\sum_{k=0}^n B_{kj} a_k = 0, \quad a_0 = 1, \quad j = 1, \dots, n. \quad (8.6.11)$$

For any arbitrary polynomial  $p \in \mathcal{L}_n^1$ , we have, in view of (8.5.2),

$$\langle f_0 - p \rangle = \langle f_0, p - f_0 \rangle + \langle f_0, f_0 \rangle = \langle f_0, f_0 \rangle,$$

and hence

$$\|f_0 - p\|^2 = \|p\|^2 - \|f_0\|^2, \quad (8.6.12)$$

which implies that  $p(z)$  has the minimal property:

$$\int_{\Gamma} |f_0(z) - p(z)|^2 ds \quad \text{is minimum in } \mathcal{L}_n^1 \text{ for } p(z) = \phi_n(z). \quad (8.6.13)$$

Again, as in §8.5,  $\|f_0 - \phi_n\| \searrow 0$  as  $n \rightarrow \infty$ , since  $\mathcal{L}_{n+1}^1 \supset \mathcal{L}_n^1$ . Then we ask, under what assumptions on  $D$  is the system of polynomials  $\{\phi_n\}$  complete in the Hilbert space  $L^2(\Gamma)$ ? An answer was given by Smirnov [1928] as

**Theorem 8.9.** *The system of polynomials in  $L^2(\Gamma)$  is complete iff the boundary  $\Gamma$  of the region  $D$  satisfies the condition*

$$\log |g'(w)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g'(r e^{i\theta})| \frac{R^2 - r^2}{R^2 - 2Rr \cos(\alpha - \theta) + r^2} d\alpha, \quad (8.6.14)$$

for all  $r < R$ , where  $z = g(w)$  is the inverse mapping function which maps the region  $|w| < R$  onto the region  $D$  in the  $z$ -plane.

A proof of this theorem can be found in Goluzin [1957:396] or [1969:449]. The condition (8.6.14) is known as the  $S$ -condition. This condition depends only on the region  $D$  and not on the normalization of  $g(w)$  or on the choice of  $a \in D$ . All such regions  $D$  whose mapping satisfies the  $S$ -condition (8.6.14) are said to belong to the *Smirnov class*  $\mathcal{S}$ . Not all regions with a rectifiable boundary belong to the class  $\mathcal{S}$ . Besides the rectifiability of  $\Gamma$ ,

however, it is sufficient for  $D \in \mathcal{S}$  if one of the following conditions is met: (i)  $D$  is convex or starlike with respect to a point in  $D$ ; (ii)  $\Gamma$  is piecewise smooth, and its smooth arcs  $\gamma_k$ ,  $k = 1, \dots, n$ , join one another with a nonzero interior angle; (iii) the ratio of the length of any arc  $\gamma_k$  of  $\Gamma = \cup \gamma_k$  to the length of its chord does not exceed a fixed limit; (iv) if  $D \notin \mathcal{S}$ , then the behavior of  $\|\phi_n\|^2$  and of  $\|f_0 - \phi_n\|^2 = \|\phi_n\|^2 - \|f_0\|^2$  is known; and (v) for any  $D$  with a rectifiable boundary  $\Gamma$ , the integral  $\int_{\Gamma} |\phi_n(z)|^2 ds \searrow 2\pi R\delta$ , where

$$\delta = \exp \left\{ \frac{1}{2\pi} \int_0^{2\pi} \log |g'(Re^{i\theta})| d\theta \right\}. \quad (8.6.15)$$

Thus, under the conditions (i), (ii), or (iii) we have  $\|f_0 - \phi_n\| \searrow 0$ , and, in view of Theorem 8.4,  $\phi_n(z) \rightarrow f_0(z)$  as  $n \rightarrow \infty$  for every region  $G \subset D$ . Hence, for  $z \in G \subset D$  the polynomial

$$\pi_{2n+1} = \int_a^z [\phi_n(\zeta)]^2 d\zeta \rightarrow f(z) \quad \text{as } n \rightarrow \infty. \quad (8.6.16)$$

Note that the polynomial  $\pi_{2n+1}(z)$  is different from the Bieberbach polynomial (8.5.18).

**Theorem 8.10.** *For the system of polynomials to be complete in a region  $D$ , it is necessary and sufficient that an arbitrary function  $F(z) \in L^2(\Gamma)$  satisfies the condition*

$$\int_{\Gamma} |F(z)|^2 ds = \sum_{k=1}^n |a_k|^2, \quad (8.6.17)$$

where  $a_k$  are the Fourier coefficients of  $F(z)$ .

PROOF. Let  $p(z)$  be an arbitrary polynomial of degree  $n$ :

$$p(z) = \sum_{k=0}^n c_k u_k(z). \quad (8.6.18)$$

Then for  $F(z) \in L^2(\Gamma)$

$$\begin{aligned} \int_{\Gamma} |F(z) - p(z)|^2 ds &= \int_{\Gamma} |F(z)|^2 ds + \int_{\Gamma} |p(z)|^2 ds - 2 \Re \left\{ \int_{\Gamma} F(z) \overline{p(z)} ds \right\} \\ &= \int_{\Gamma} |F(z)|^2 ds + \sum_{k=0}^n |c_k|^2 - 2 \Re \left\{ \sum_{k=0}^{\infty} a_k c_k \right\} \\ &= \int_{\Gamma} |F(z)|^2 ds - \sum_{k=0}^n |c_k|^2 + \sum_{k=0}^n |a_k - c_k|^2. \end{aligned} \quad (8.6.19)$$

Thus, the polynomial  $p(z)$  defined by (8.6.18) with  $c_k = a_k$  attains the minimum of the integral

$$\int_{\Gamma} |F(z) - p(z)|^2 ds, \quad (8.6.20)$$

and, in view of (8.6.18), this minimum yields

$$\int_{\Gamma} |F(z)|^2 ds - \sum_{k=0}^n |c_k|^2 = 0. \quad (8.6.21)$$

Hence, a system of polynomials is complete iff the difference (8.6.21) approaches zero as  $n \rightarrow \infty$  for an arbitrary function  $F \in L^2(\Gamma)$ . ■

**Theorem 8.11.** *If the  $S$ -condition (8.6.14) is satisfied, then an arbitrary function  $F(z) \in L^2(\Gamma)$  can be represented in  $D$  by a Fourier series*

$$F(z) = \sum_{k=0}^{\infty} a_k u_k(z), \quad (8.6.22)$$

which converges uniformly in  $D$ , where

$$a_k = \int_{\Gamma} f(w) \overline{u_k(w)} ds. \quad (8.6.23)$$

PROOF. Since the minimum of the integral (8.6.20) is attained out of all polynomials  $p_n(z)$  by a polynomial defined by (8.6.18) with  $c_k = a_k$ ,  $k = 0, 1, \dots, n$ , then, if the  $S$ -condition is satisfied, we have

$$\lim_{n \rightarrow \infty} \int_{\Gamma} |F(\zeta) - p_n(\zeta)|^2 ds = 0. \quad (8.6.24)$$

But by Cauchy's formula in  $D$

$$F(z) - p_n(z) = \frac{1}{2i\pi} \int_{\Gamma} \frac{F(\zeta) - p_n(\zeta)}{\zeta - z} d\zeta.$$

If  $\bar{G}$  is a closed subset of  $D$  and  $\delta$  is the distance from  $G$  to  $\Gamma$ , then for  $z \in \bar{G}$

$$\begin{aligned} |F(z) - p_n(z)| &\leq \frac{1}{2\pi\delta} \int_{\Gamma} |F(\zeta) - p_n(\zeta)| ds \\ &\leq \frac{1}{2\pi\delta} \sqrt{l \int_{\Gamma} |F(\zeta) - p_n(\zeta)|^2 ds}, \end{aligned} \quad (8.6.25)$$

where  $l$  is the length of  $\Gamma$ , and this, in view of (8.6.24), implies that  $p_n(z) \rightarrow F(z)$  uniformly on  $\bar{G}$  as  $n \rightarrow \infty$ . But  $p_n(z)$  is a finite part of the Fourier series (8.6.22), which proves the theorem. ■

**Example 8.3.** Let  $E$  denote the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ . The Bergman kernel for  $E$  has the form

$$K(z, a) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(n+1)U_n(z)\overline{U_n(a)}}{\rho^{n+1} - \rho^{-n-1}}, \quad \rho = (a+b)^2,$$

where  $U_n(z) = (1 - z^2)^{-1/2} \sin((n+1) \cos^{-1} z)$  are the Chebyshev polynomials of the second kind and degree  $n$  (Nehari [1952: 258-259]). *bull*

**Example 8.4.** Let the arc  $\Gamma$  be defined by the ellipse  $E: \frac{(x-x_c)^2}{a^2} + \frac{(y-y_c)^2}{b^2} = 1$ ,  $a > b$ , and let the parametric equation of  $\Gamma$  be  $z = \gamma(s) = z_c + ae \cos(s - iq)$ ,  $0 \leq s_1 < s < s_2 < 2\pi$ , where  $z_c = x_c + iy_c$  is the center  $C$ ,  $e = \sqrt{1 - b^2/a^2}$  the eccentricity of the ellipse,  $\cosh q = 1/e$ , and  $s_2 - s_1 < 2\pi$ . Then the function  $z = \gamma(\zeta)$ ,  $\zeta = s + it$ , is univalent

in the strip  $\{\zeta : \zeta + s + it, s_1 < s < s_2, -\infty < t < q\}$ , and the region  $G^*$  is a symmetric subregion of the rectangle  $\{\zeta : \zeta = s + it, s_1 < s < s_2, -q < t < q\}$  (Papamichael, Warby and Hough [1983: 157]). *bull*

## 8.7 Numerical Parameter Problem

The Schwarz-Christoffel transformation defined by (7.1.7) or (7.1.15) is

$$z(w) = C_1 \int_{w_0}^w \prod_{j=1}^n (t - w_j)^{-\alpha_j} dt + C_2. \quad (8.7.1)$$

The problem is to determine all  $w_j$  and the constants  $C_1$  and  $C_2$ . There are different methods to evaluate these parameters, e.g., those described in von Koppenfels and Stallman [1959], Kantorovich and Krylov [1964], Gaier [1964], and Henrici [1986: vol. 3, ch. 16], and Bjørstad and Grusse [1987]. We will use the following four methods:

**8.7.1 Parameter Method.** The parameters  $w_j$ ,  $j = 1, 2, \dots, n$  and  $C_1, C_2$  are determined by using the boundary conditions, i.e., by equating values of  $z(w)$  in the  $z$ -plane to the values of the integral evaluated at the corresponding points  $w_j$  on the  $v$ -axis. Although certain cases have been determined exactly in Chapter 6, numeral methods are needed when  $n$  is large. First, we assume that we have reduced the problem such that  $z(0) = 0$ , which yields  $C_2 = 0$ . Next, let the distance between two adjacent vertices  $z_k$  and  $z_l$  of the polygon be denoted by  $\Delta z_{kl}$  and the corresponding values determined from integration by  $\Delta F_{kl}$ , i.e.,

$$\Delta z_{kl} = |z_l - z_k|; \quad \Delta F_{kl} = C_1 \int_{z_k}^{z_l} \prod_{j=1}^n (t - w_j)^{-\alpha_j} dt. \quad (8.7.2)$$

Now, let the next vertex be  $z_m$ . Then equating the ratios of the lengths  $\Delta z_{kl}$  and  $\Delta z_{lm}$  we get

$$\frac{\Delta z_{kl}}{\Delta z_{mk}} = \frac{\Delta F_{kl}}{\Delta F_{mk}}. \quad (8.7.3)$$

This equation results in a set of simultaneous, nonlinear equations which can be solved using the least-error-squared condition. For example, if the error is denoted as  $e = \Delta z - \Delta F$ , we can minimize the sum of all the  $e^2$ . Powell [1964] provided the method of conjugate direction, which was used by Lawrenson and Gupta [1968] to develop a minimization algorithm, who found that displacement of limits in the order of 0.001 and 0.005 were sufficient to get accurate results in small intervals.

**8.7.2 van Dyke's Method.** (van Dyke [1975: 96]) This method deals with the following situation: Since a vertex, say  $z_j$ , of the polygon is mapped onto the point  $w_j$  on the  $u$ -axis, each finite  $w_j$  contributes a term  $(t - w_j)^{-\alpha_j}$  in the Schwarz-Christoffel transformation (8.7.1). If an approximate, yet almost accurate, solution is required in the interval  $w_j^- < w < w_j^+$ , van Dyke's method works with accuracy of the order of 0.001 to 0.005 as determined by Lawrenson and Gupta [1968] and Binns [1971]. This method uses the fact that the factor  $(t - w_j)^{-\alpha_j}$  in (8.7.1) predominates as the singular point  $w_j$  is approached by  $w$ , while the other factors  $(t - w_m)^{-\alpha_m}$ ,  $m \neq j$ , remain almost constant in their magnitude and direction as integration is taken from  $w_m^-$  to  $w_m^+$ . Thus, the difference

$$\Delta z_m = z_m^+ - z_m^- = C_1 \int_{w_m^-}^{w_m^+} \prod_{j=1}^n (t - w_j)^{-\alpha_j} dt \quad (8.7.4)$$

changes to

$$\Delta z_m = C_1 \left\{ \prod_{m \neq j} (w_m - w_j)^{-\alpha_j} \right\} \frac{(w - w_m)^{1-\alpha_m}}{1 - \alpha_m} \Bigg|_{w_m^-}^{w_m^+}. \quad (8.7.5)$$

However, this method will fail in the following cases and should be avoided:

(i) The factor  $1 - \alpha_m$  (in the denominator in (8.7.5)) will remain nonnegative since any vertex with  $|\alpha_m| > 1$  occurring at infinity in the  $z$ -plane will be dropped out from the integral. The difficulty would arise only when this factor itself becomes infinitely large when  $\alpha_m = 1$ . Such situation normally arises in idealized representations, e.g., of capacitor plates or air foils of zero thickness where the exterior angles of  $\pi$  as the boundaries are traversed in the counter-clockwise direction. However, extremely narrow slits are acceptable since they have  $\alpha = -\pi$ .

(ii) This method will give inaccurate results in the case when the singular points are dense on the  $u$ -axis. This situation may require appropriate scaling and placements of such points.

(iii) If the images of the vertices in the  $z$ -plane occur on the  $u$ -axis at infinity, the integration in (8.7.4) will be taken between infinitely large intervals of  $w$ . Imagine the case when a vertex in the  $z$ -plane is at infinity. Then the open polygon can be transformed into a symmetric, closed polygon, as seen in Figure 8.6, due to Davis [1979].

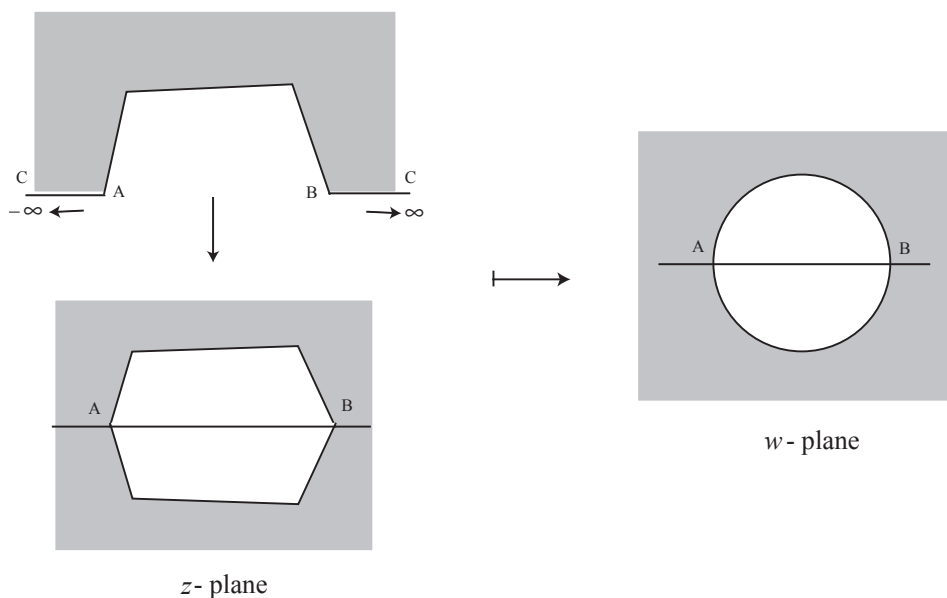


Figure 8.6 Use of symmetry to avoid vertices at infinity.

**8.7.3 Trefethen's Method.** (Trefethen [1980]) Instead of mapping the  $z$ -plane onto the upper-half  $w$ -plane, this method is based on the mapping of the polygon in the  $z$ -plane onto the unit disk  $|w| \leq 1$  (see §7.1.3), and this is the advantage over other methods. Thus, all vertices  $z_j$  of the polygon are mapped onto the unit circle. For example, a vertex  $z_j$  will correspond to  $w_j = e^{i\Theta_j}$ , thereby  $\Theta_j$  will be the only parameter to be determined. For an efficient integration of Eq (8.7.1) the endpoints  $w_j$  should not be too close to singular points which are images of the other vertices. To achieve it, a careful spacing of the angles



$\Theta_{j-1}, \Theta_j, \Theta_{j+1}$  should be maintained while retaining their power sequence. In the words of Trefethen, “no singularity shall lie closer to an interval of integration than half the length of that interval.” A computer program SCPACK, prepared by Trefethen, evaluates the integrals using Gauss-Jacobi quadrature, and the nonlinear equations (8.7.5) are solved using an iterative procedure that is based on an initial guess and subsequent corrections using Powell’s minimization method (Fletcher and Powell [1963], Powell [1964]).

Trefethen has solved some practical problems, like flat polygonal registers (Trefethen [1984]), and Hall generator elements which involves oblique derivative boundary conditions (Trefethen and Williams [1986]). Later, Elcrat and Trefethen [1986] extended this method to open polygons in free-streamline flow problems without using the log-hodograph method (see §8.4 for minimum area problem).

Dias [1986] has applied this method to dividing-flows and point sources (singularities) such as wells. However, instead of using the Gauss-Jacobi quadrature, he has used change of variable to remove singularities, and thus, deals efficiently with vertices at infinity.

**8.7.4 Foster-Anderson’s Method.** (Foster and Anderson [1974]) This method uses the elliptic functions to numerically evaluate the Schwarz-Christoffel integral.

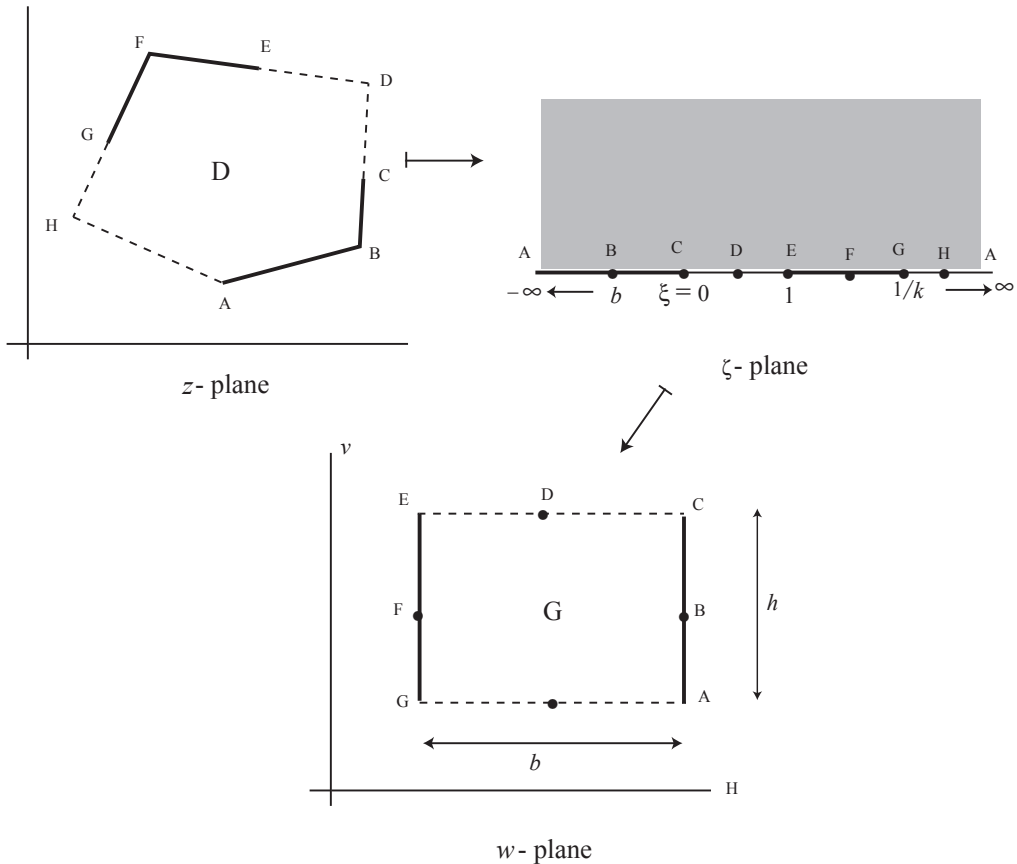


Figure 8.7 Synthesis of a polygon to conform with a desired modulus.

The algorithm for this method has been published by Warner and Anderson [1981], and

its details are as follows: Consider a closed polygon with eight vertices in the  $z$ -plane, marked A through H in Figure 8.7, enclosing the region  $D$ . The boundary  $\partial D$  together with the eight vertices is mapped onto the real axis  $\xi$  in the  $\zeta$ -plane ( $\zeta = \xi + i\eta$ ), such that the vertices C, E, and A are mapped onto  $\xi_C = 0$ ,  $\xi_E = 1$ , and  $\xi_A = \infty$ , respectively. However, we do not have any information about the mapping of the vertices B, D, F, G and H, except that G is determined for the case when the  $\zeta$ -plane is mapped onto the rectangle ABCD in the  $w$ -plane, where the segments ABC and EFG correspond to opposite sides. This situation can be compared to the case of a conducting plane  $D$  in which the segments with heavy lines represent electrodes attached to the plate, and with the dashed lines represent the border, or lines of symmetry, through which no current passes. Then the ratio of the height and base of the rectangle can be defined as a nondimensional modulus by

$$\text{mod} = \frac{h}{b} = \frac{\text{conductance}}{\sigma\tau} = \frac{K'(k)}{K(k)}, \quad (8.7.6)$$

where  $\sigma$  denotes the conductivity and  $\tau$  the thickness of the plate, and  $K'$  and  $K$  are elliptic functions (see Appendix F). In the case when a square is required in the  $w$ -plane, just set  $h = b$ , and the modulus will be 1.

According to Foster and Anderson [1974], there are two phases of the problem: one analytical and the other synthetic. In the former phase one should find the point  $\xi_G = 1/k$  such that the boundary  $\partial D$  maps onto the boundary of the rectangle in the  $w$ -plane. In the other phase, one should find a specified ratio of  $K'(k)/K(k)$  that is proportional to conductance, or some other topologically determined physical parameter, so that  $\xi_G$  is located properly. In both cases the modulus  $k$  of the elliptic functions must be evaluated since  $\xi_G = 1/k$ . This is accomplished as follows:

The value of the modulus  $k$  can be set by specifying  $h/b$ , thus  $\xi_G$  is fixed. Then proceed to locate the remaining vertices B, D, F, and H on the  $xi$ -axis. Foster and Anderson [1974] used the method of rearranging the terms of the Schwarz-Christoffel integral and approximating of the integral by factoring out vertex pairs of the form  $\{(p - \xi_l)(\xi_m - p)\}^{-1/2}$ , thereby leaving behind function  $f(p)$ , where  $p$  is the variable of integration (same as  $\xi$ ). Next, an approximation is given by

$$\begin{aligned} \Delta z_m &= C \int_{\xi_l}^{\xi_m} \prod_{j=1}^n (t - \xi_j)^{-\alpha_j} dt = C \int_{\xi_l}^{\xi_m} \frac{f(p)}{\{(p - \xi_l)(\xi_m - p)\}^{1/2}} dp \\ &= C \frac{\pi}{N} \sum_{j=1}^N f(p_j), \end{aligned} \quad (8.7.7)$$

where

$$p_j = \frac{\xi_m + \xi_l}{2} + \frac{\xi_m - \xi_l}{2} \cos\left(\frac{(2j-1)\pi}{2N}\right),$$

and  $n$  is the number of vertices. Note that  $N$  denotes the number of elements in the series expansion, and the results are more accurate, the larger  $N$  is. The series expansion is available in Abramowitz and Stegun [1972: 889].

Only  $C$  and  $\xi_1, \dots, \xi_n$  are known so far, as can be determined using (8.7.1)-(8.7.3). Thus, we get

$$|z_m - z_j| = C \frac{\pi}{N} \sum_{j=1}^N f(p_j^{lm}),$$



For example, consider the case for which  $K'/K = 1.28$  holds for the trapezoidal segment which represents  $1/8$  of the region in the  $z$ -plane. This fixed  $k = 0.5$ , and thus,  $\xi_A = 1/k^2 = 4$ . With this value of  $\xi_A$ , integrals (8.7.10) and (8.7.11) give  $V_1 = 2.3 = V_3$ , and  $a = 0.5$ . ■

For other examples with different cases with more vertices are available in Warner and Anderson [1981], and Schinzinger and Laura [2003: 293].

**8.7.5 Unit Disk onto Polygon.** (Subroutine WSC) To evaluate the inverse map 7.4 for a given point  $z$  in the unit disk or on the circle, we compute the integral

$$w = w_0 + C \int_{z_0}^z \prod_{j=1}^N \left(1 - \frac{z'}{z_j}\right)^{-\alpha_j} dz', \quad (8.7.12)$$

where  $w_0 = w(z_0)$ , and the endpoint  $z_0$  may be a point in the closed disk at which the image  $w(z_0)$  is known and finite. There are three possible choices for the point  $z_0$ : (i)  $z_0 = 0$ , which gives  $w_0 = w_c$ ; (ii)  $z_0 = z_k$  for some  $k$ , which gives  $w_0 = w_k$  which is vertex of the polygon  $P$ ; and (iii)  $z_0$  is some other point in the disk at which  $w$  has already been computed. Note that in case (i) and (iii), neither endpoint has a singularity, and the evaluation of (8.7.12) can be done using compound Gauss quadrature. However, in case (ii) a singularity of the form  $(1 - z/z_k)^{-\alpha_k}$  is present at one of the endpoints and the other point has no singularity. Hence, the best way to compute  $w(z)$  is: If  $z$  is close to a singular point  $z_k$  (but not the one with  $w_k = \infty$ ), use subroutine ZQUAD; otherwise use SCFUN. In either case use compound Gauss-Jacobi quadrature, taking the same number of nodes as was used in solving the parameter problem. This procedure will evaluate  $w(z)$  with complete accuracy by computing all accessory parameters, where quadrature nodes and weights need only be computed once.

**8.7.6 Polygon onto Unit Disk.** (Subroutine ZSC). To compute the inverse map  $z = z(w)$  there are at least two methods: (i) The most direct approach is to regard the direct map  $w = w(z)$  as a nonlinear equation and solve it for  $z$ , given some fixed value  $w$ . The solution may then be found iteratively by Newton's method or a related method, where  $w(z)$  must be evaluated at each step using the compound Gauss-Jacobi quadrature along a straight line segment whose initial point is kept fixed throughout the iteration.

(ii) The second method is to invert the Schwarz-Christoffel transformation formula (7.1.4) to yield

$$\frac{dz}{dw} = \frac{1}{C} \prod_{k=1}^N \left(1 - \frac{w}{w_k}\right)^{\alpha_k}. \quad (8.7.13)$$

The inversion is possible because  $w = w(z)$  is a conformal mapping, which means that  $|dw/dz| > 0$  everywhere. Eq (8.7.13) may also be regarded as an ordinary differential equation of the form  $\frac{dz}{dw} = g(w, z)$  in one complex variable  $w$ . If a pair of values  $(z_0, w_0)$  is known and the new values  $z(w)$  is sought, then  $z$  may be computed using a numerical o.d.e. solver, where the path of integration is taken to be any curve from  $w_0$  to  $w$  which lies within the polygon  $P$ .

Trefethen [1979] has combined both of these methods, such that the second method is first used to generate an initial estimate to be used in the first. Thus, the o.d.e formulation begins, first by expressing Eq (8.7.13) as a system of first-order o.d.e. in two real variables, and then using the code ODE by Shampine and Allen [1973], and integrating along straight line segment from  $w_c$  to  $w$ , whenever possible. Since the polygon  $P$  is not convex, more

than one line segment may be required to connect  $w_0$  and  $w$ . Note that it will not do to take  $w_0 = w_k$  for some vertex  $w_k$  without special care, because Eq (8.7.13) is singular at  $w_k$ . Thus, the following steps are needed to compute  $z = z(w)$  rapidly to full accuracy:

STEP 1. Solve Eq (8.7.13) using the ODE package whenever possible along a line segment from  $w_c$  to  $w$ ; call the result  $\tilde{z}$ ; and

STEP 2. Solve the equation  $w(z) = w$  for  $z$  by Newton's method, using  $\tilde{z}$  as an initial guess.

Fortran programs of all subroutines are available in Trefethen [1979: 48-55].

## 8.8 Generalized Schwarz-Christoffel Parameter Problem

Trefethen and Williams [1986] have used Schwarz-Christoffel transformations to numerically solve Laplace's equation in a polygonal domain with boundary conditions that the derivative in the direction at a specified oblique angle from the normal is zero along each side of the polygon. The solution is constructed by taking the real part of the mapping function that maps the physical domain onto another (model) domain with straight sides according to the angles given in the boundary conditions.

Given a set of real numbers  $\theta_k$ ,  $k = 1, 2, \dots, n$ , let  $u_n(z) = \frac{\partial u}{\partial n}(z)$  denote the inward normal derivative of a function  $u$ , and let  $u_\sigma = \frac{\partial u}{\partial \sigma}$  be its tangential derivative along the side  $\Gamma_k$  in the positive sense, of a polygon  $D$  with vertices  $z_k$  for each  $k = 1, 2, \dots, n$ . Then the problem considered in the above work, known as the *oblique derivative problem*, is as follows:

**Problem O.** Find the functions  $u \in C^2(D) \cap C^1(D \cup \bigcup_k \Gamma_k)$  which satisfy Laplace's equation  $\nabla^2 u(z) = 0$ ,  $z \in D$ , subject to the homogeneous oblique boundary conditions

$$\cos \theta_k u_n(z) + \sin \theta_k u_\sigma(z) = 0. \quad (8.8.1)$$

This problem is illustrated in Figure 8.7(a). If  $\theta_k = 0 \pmod{\pi}$ , then we have a Neumann condition on the side  $\Gamma_k$ , while  $\theta_k = \pi/2 \pmod{\pi}$  gives the tangential condition.

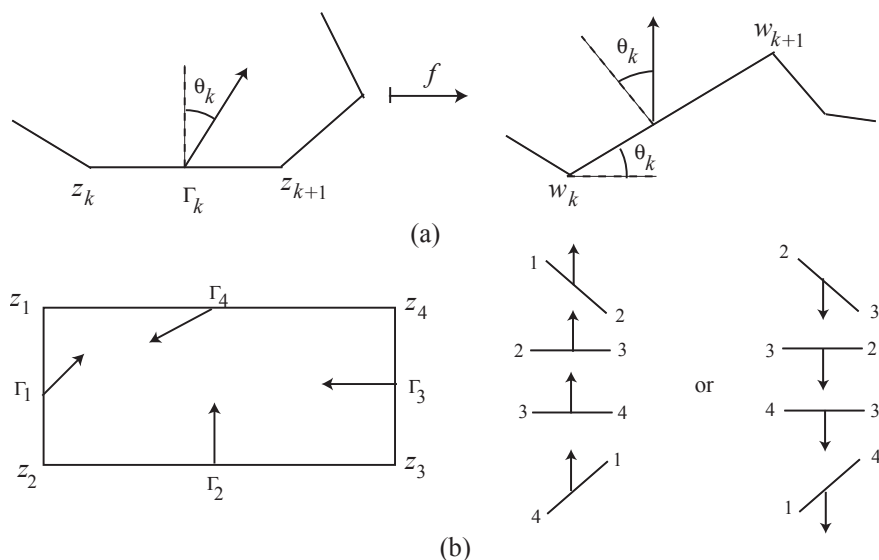


Figure 8.9 Problem O.

This problem for a rectangle bounded by four sides  $\Gamma_1, \dots, \Gamma_4$ , is presented in Figure 8.9(b); the Neumann condition holds on  $\Gamma_1$  and  $\Gamma_4$ , oblique derivative conditions at  $45^\circ$  from the normal in the indicated directions. To apply the conformal mapping method, we must find a polygonal image domain  $f(D)$  whose sides are oriented as in Figure 8.9 (right-side figure).

The conformal mappings begin in the following order: First, as shown in Figure 8.10, the problem from  $D$  onto the upper half  $t$ -plane by a conformal mapping  $\phi$  with  $\phi(z_k) = \infty$  for some vertex  $z_k$ . It is known that  $\phi$  must be the Schwarz-Christoffel mapping

$$\phi^{-1}(t) = C_1 + C_2 \int_{T_\phi}^t \prod_{\substack{k=1 \\ k \neq k_\infty}}^n (t' - t_k)^{-\alpha_k} dt', \quad (8.8.2)$$

where  $C_1, C_2 \in \mathbb{C}$ , and  $T_\infty \in \mathbb{R}$  are constants;  $t_k = \phi(z_k)$  are the prevertices on the real axis and  $\alpha_k \pi$  is the external angle at the vertex  $z_k$ ,  $-1 \leq \alpha_k \leq 3$ . However, the difficulty is that the values of  $\{t_k\}$  are not known in advance. Thus, determining  $\phi$  numerically requires an iterative method to determine these values. The Fortran package SCPACK (§24.10) is used for this purpose.

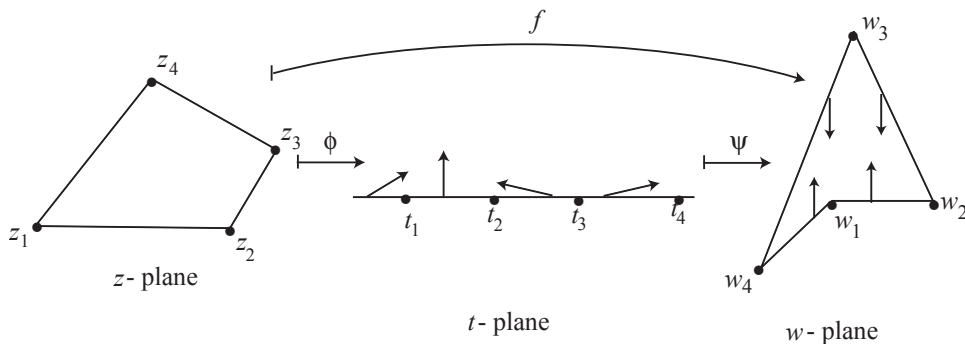


Figure 8.10 Schwarz-Christoffel transformations.

The next step is to map the upper half  $t$ -plane onto  $f(D)$  by a function  $\psi(t)$ , as shown in Figure 8.10. This is again another Schwarz-Christoffel transformation involving the same prevertices  $t_k$  and new angle parameters  $\alpha'_k$ , and other new features. The function  $\psi$  may have branch points in the upper half-plane; to handle them, additional factors  $(t' - s_k)(t' - \bar{s}_k)$ ,  $s_k \in \mathbb{R}$  are used; also the factors  $(t' - s_k)$  are included whenever there are points along some  $\Gamma_k$  where  $f'(z) = 0$ . Finally, since  $f(D)$  is not required to be embeddable in the plane, the parameters  $\alpha'_k$  need not lie in  $[-1, 3]$ . Hence, the general map  $\psi$  with algebraic singularities can be written as

$$\psi(t) = C_3 + e^{i\theta_{k_m-1}} \int_{T_\psi}^t p(t') \prod_{\substack{k=1 \\ k \neq k_\infty}}^n (t' - t_k)^{m_k - \alpha'_k} dt', \quad \alpha'_k = \frac{1}{\pi} (\theta_k - \theta_{k-1}), \quad (8.8.3)$$

where  $p(t')$  is a polynomial with real coefficients, each  $m_k$  is an integer,  $T_\psi$  is any point in  $(t_{k-1}, \infty)$ , and  $C_3$  is a complex constant.

This is a *generalized Schwarz-Christoffel parameter problem*, where  $p$  is linear, which makes it easily solvable. The idea of generalized parameter problems was introduced by

Trefethen. Boundary value problems for elliptic equations in non-smooth domains, especially Poisson's equation on polygonal domains with oblique derivative, have been studied, e.g., by Grisvard [1980], and Kondrat'ev and Olcinin [1983].

## 8.9 Proofs

To prove the Schwarz-Christoffel formula (7.1.2), which leads to Eq (8.1.1), we will follow the argument from Driscoll and Trefethen [2002], and prove

**Theorem 8.12.** *Let  $\Gamma$  be a polygon with vertices  $w_1, \dots, w_n$  and exterior angles  $\alpha_1\pi, \dots, \alpha_n\pi$  in counterclockwise order. Let  $w = f(z)$  be any conformal bijective map from the upper half-plane  $\Im\{z\} > 0$  onto  $G = \text{Int}(\Gamma)$  in the  $w$ -plane such that  $w(\infty) = w_n$ . Then the mapping  $w$  can be written in the form (7.1.2), where  $z_1 < z_2 < \dots < z_n$  are real numbers satisfying  $w(z_k) = w_k$  for  $k = 1, 2, \dots, n$ .*

Note that Eq (7.1.2) involves improper contour integrals.

**Lemma 8.1.** *Let  $\Gamma$  be a polygon and let  $f$  map the upper half-plane  $H \equiv \Im\{z\} > 0$  conformally onto  $G$ . Then  $f$  extends continuously to the closure  $\bar{H}$ . More precisely, there exists a homeomorphism  $F$  from  $\bar{H}$  onto  $\Gamma \cup G$  satisfying  $F|_H = f$ .*

PROOF. Consider a vertex  $a = w_k$  of  $\Gamma$ . Then the function  $z \mapsto z^{1/(1-\alpha_k)}$  makes the angle straight between the two edges on which  $a$  lies. Thus,  $g = f^{-1}$  extends to a continuous function  $\phi$  on all of  $G \cup \Gamma$ . Since  $\phi'$  does not vanish on any open edge of  $\Gamma$ , then, by mean-value theorem,  $\phi$  is injective on each closed edge. In particular, it cannot reverse direction on adjacent edges and thus,  $\phi|_\Gamma$  is injective. Since  $g' \neq 0$  on  $G$ , by the open mapping theorem we have  $\phi(\Gamma) \cap \phi(G) = \emptyset$ , which means that  $\phi$  is injective on all  $G \cup \Gamma$ . Since  $\Gamma$  is a closed curve,  $\phi(\Gamma)$  is defined on the extended real axis. The proof is completed by taking  $F = \phi^{-1}$ . ■

**Lemma 8.2.** *Let  $f, \Gamma$ , and  $G$  be fixed, and let  $a \in \Gamma$  be not a vertex. Then the function  $g$  extends to an analytic function  $\phi$  in a disk centered at  $a$  such that  $g(a) \neq 0$ .*

PROOF OF THEOREM 8.12. In view of Lemma 8.1, the map  $w$  can be continuously extended to the closed upper half-plane. For  $k = 1, 2, \dots, n$ , denote the prevertices by  $z_k = w^{-1}(w - k)$ . Using the Schwarz reflection principle (§2.6.6) we can extend  $w$  and  $w'$  analytically across the real axis everywhere except the prevertices. According to Lemma 8.1, the function  $z \mapsto (z - z_k)^{-\alpha_k} w'(z)$  has an analytic extension in the neighborhood of  $z_k$ . Thus, we have

$$w'(z) = (z - z_k)^{-\alpha_k} \psi(z),$$

where  $\psi(z)$  is some function analytic in the neighborhood of  $z_k$ . Hence, we get

$$\begin{aligned} w'(z) &= (z - z_k)^{-\alpha_k} \psi(z), \\ w''(z) &= -\alpha_k (z - z_k)^{-\alpha_k - 1} \psi(z) + \psi'(z) (z - z_k)^{\alpha_k}, \end{aligned}$$

so that

$$\frac{w''(z)}{w'(z)} = \frac{-\alpha_k (z - z_k)^{-\alpha_k - 1}}{\psi(z)} (z - z_k)^{\alpha_k} \psi(z) + \frac{\psi'(z) (z - z_k)^{-\alpha_k}}{(z - z_k)^{-\alpha_k} \psi(z)} = -\frac{\alpha_k}{z - z_k} + \frac{\psi'(z)}{\psi(z)},$$

which implies that  $\frac{w''(z)}{w'(z)}$  has a simple pole at  $z = z_k$  with residue  $-\alpha_k$ . Hence, the function  $\frac{w''(z)}{w'(z)} + \sum_{k=1}^n \frac{\alpha_k}{z - z_k}$  is an entire function. Also, since all the prevertices are finite,

$w$  is analytic at  $z = \infty$ . Using the Laurant series expansion we find that  $\frac{w''(z)}{w'(z)} \rightarrow 0$  as  $z \rightarrow \infty$ , such that each term of the summand goes to zero as  $z \rightarrow \infty$ , which means that it is a bounded entire function. Then by Liouville's theorem (§2.4) the expression  $\frac{w''(z)}{w'(z)}$  is constant and is identically zero. Next, since  $(\log(w'))' = \frac{w''(z)}{w'(z)}$ , we get

$$\begin{aligned} (\log(w'))' &= \sum_{k=1}^n \frac{-\alpha_k}{z - z_k}, \\ \log(w') &= \int \sum_{k=1}^n \frac{-\alpha_k}{z - z_k}, \\ &= \sum_{k=1}^n \int \frac{-\alpha_k}{z - z_k} = -\sum_{k=1}^n \alpha_k \ln(z - z_k) + C_1 = \sum_{k=1}^n \ln(z - z_k)^{-\alpha_k} + C_1 \\ w' &= \exp \left\{ \sum_{k=1}^n \ln(z - z_k)^{-\alpha_k} + C_1 \right\} = \exp \left\{ \sum_{k=1}^n \ln(z - z_k)^{-\alpha_k} \right\} \exp(C_1) \\ &= C \prod_{k=1}^n \exp \left\{ \ln(z - z_k)^{-\alpha_k} \right\}, \quad \text{where } C = \exp \{C_1\} = C \prod_{k=1}^n (z - z_k)^{-\alpha_k} \end{aligned}$$

which after integration yields

$$w = C_1 \int \prod_{k=1}^n (z - z_k)^{-\alpha_k} + C_2. \blacksquare$$

NOTES. cs442.nb:

```
A[j_, k_] := Integrate[ Integrate[ (x+ I*y)^j * (x-I*y)^k,
{x, -1,1}], {y, -1,1}];
MatA = Table[A[j,k], {j,1,8}, {k,1,8}];
MatrixForm[MatA];
B=Table[A[j,0], {j,1,8}];
c=LinearSolve[MatA,- B];
(* These are the coefficients of phi8[z] *)
phi8[z_] := 1 + c . Table[z^i, {i, 8}];
phi8[z];
(* The mapping function is given by f'[z]=phi8[z] *)
f[z_] := Integrate[phi8[t], {t, 0,z}];
f[z]
```

REFERENCES USED: Ahlfors [1966], Birkoff, Young and Zarantonello [1951], Boas [1987], Carrier, Krook and Pearson [1966], Elcrat [1982], Driscoll and Trefethen [2002], Elcrat and Trefethen [1986], Foster and Anderson [1974], Gaier [1964], Goluzin [1957; 1969], Kantorovich and Krylov [1958], Kythe [1998], Nehari [1952], Papamichael, Warby and Hough [1983]. Robertson [1965], Schinzingler and Laura [2003], Trefethen [1980], Trefethen and Williams [1986], von Koppenfels and Stallmann [1959], Warner and Anderson [1981]